Submittal Review Response

Project Name:	***Hilo WWTP Rehabilitation and Replacement Project Phase 1***
Submittal No.:	***15230-002.1***
Date:	*9/26/2026*

Client:	County of Hawaiʻi	**Carollo Project No.:**	203975
Contractor:	Nan, Inc.		
Submittal Name:	PVC SCH80		
Reviewed By:	Adria Nyarko, Adrienne Fung		

SUBMITTAL REVIEW		
Review is for general compliance with contract documents. No responsibility is assumed by Carollo for correctness of quantities, dimensions, and details. No deviation or variation is approved unless specifically addressed in these review comments. Refer to Section 01330 for additional requirements. The Contractor shall assume full responsibility for coordination with all other trades and deviations from contract requirements.		
Approved	☐	No Exceptions
	☒	Make Corrections Noted - See Comments
	☐	Make Corrections Noted - Confirm
Not Approved	☐	Correct and Resubmit
	☐	Rejected - See Remarks
Receipt Acknowledged	☐	Filed for Record
	☐	With Comments - Resubmit

Review Comments:

1. The marked-up specification pages do not include the submittal requirement page. This submittal includes product data but not shop drawings. It is unclear if shop drawings required per Section 15230-1.04.C will be provided in a separate submittal. Please confirm. Also note that the Contractor shall include the entire marked up specification section for all future submittals per Section 01330-1.03.G.1.

    Contractor's Response: Incorporated

    Engineer's Response: Acknowledged. Shop drawings to be submitted later.

2. The submittal states that fittings are in accordance with ASTM D2467. Confirm that pressure fittings are in accordance with ASTM D2467, and DWV fittings will be in accordance with ASTM D2665 per Section 15230-2.02.A.2.b and c.

    Contractor's Response: The vendor confirmed pressure fittings are in accordance with ASTM D2467. Also stated, "DWV" note is not applicable—DWV fittings are generally not used with SCH 80 pipe

    Engineer's Response: Acknowledged.

3. The submittal states that the material cell classification is 12454. The cell classification should be 12454-B per Section15230-2.02.A.1.a..

    Contractor's Response: See attachment from vendor stating that "B" for cell classification is obsolete

Engineer's Response: Acknowledged.

4. It is unclear which ASTM standard will be followed for product marking. Confirm that markings will be in accordance with ASTM D1785 per Section 15230-2.08.A.1.

    Contractor's Response: Incorporated – highlighted in "product marking section

    Engineer's Response: Please confirm that markings will be in accordance with ASTM D1785 per Section 15230-2.08.A.1. The standard is not highlighted in the submittal.

5. It is unclear which ASTM standard will be followed for solvent weld joint installation. Confirm that installation will be in accordance with ASTM D2855 per Section 15230-3.01.B.1.

    Contractor's Response: Incorporated – highlighted in "product marking section"

    Engineer's Response: Acknowledged.

☑ High Priority

CONTRACTOR SUBMITTAL TRANSMITTAL FORM REV. A

Owner:	County of Hawaii		
Contractor:	Nan, Inc.	**Project No.:**	WW-4705R
Project Name:	Hilo WWTP Phase 1	**Submittal Number:**	15230-002.1
Submittal Title:	PVC SCH80	☐ For Information Only	
TO:	County		
From:	Nan Inc.		

Specification No. and Subject of Submittal / Equipment Supplier			
Spec: 15230	**Paragraph:** 1.04		
Authored By: Core & Main		**Date Submitted:**	09/11/2025

Submittal Certification

Check Either (A) or (B):

☐ (A) We have verified that the equipment or material contained in this submittal meets all the requirements specified in the project manual or shown on the contract drawings with no exceptions.

☑ (B) We have verified that the equipment or material contained in this submittal meets all the requirements specified in the project manual or shown on the contract drawings except for the deviations listed.

Certification Statement: By this submittal, I hereby represent that I have determined and verified all field measurements, field construction criteria, materials, dimensions, catalog numbers and similar data, and I have checked and coordinated each item with other applicable approved shop drawings and all Contract requirements.

General Contractor's Reviewer's Signature: Nitesh. N

Printed Name and Title: Nitesh Nakirikanti, Project Engineer

In the event, Contractor believes the Submittal response does or will cause a change to the requirements of the Contract, Contractor shall immediately give written notice stating that Contractor considers the response to be a Change Order.

Firm: **Signature:** **Date Returned:**

PM/CM Office Use

Date Received GC to PM/CM:

Date Received PM/CM to Reviewer:

Date Received Reviewer to PM/CM:

Date Sent PM/CM to GC:

Nan, Inc

PROJECT: HILO WWTP REHABILITATION AND REPLACEMENT PROJECT - PHASE 1

JOB NO. WW-4705R

THIS SUBMITTAL HAS BEEN CHECKED BY THIS CONTRACTOR. IT IS CERTIFIED CORRECT, COMPLETE, AND IN COMPLIANCE WITH CONTRACT DRAWINGS AND SPECIFICATIONS. ALL AFFECTED CONTRACTORS AND SUPPLIERS ARE AWARE OF, AND WILL INTEGRATE THIS SUBMITTAL (UPON APPROVAL) INTO THEIR OWN WORK.

DATE RECEIVED 9/11/2025
SPECIFICATION SECTION # 15230
SPECIFICATION 1.04
PARAGRAPH 1.04
DRAWING N/A
SUBCONTRACTOR N/A
SUPPLIER Core & Main
MANUFACTURER N/A

CERTIFIED BY CQCM or Designee : [signature]

Review Comments:

1. The marked-up specification pages do not include the submittal requirement page. This submittal includes product data but not shop drawings. It is unclear if shop drawings required per Section 15230-1.04.C will be provided in a separate submittal. Please confirm. Also note that the Contractor shall include the entire marked up specification section for all future submittals per Section 01330-1.03.G.1.

   Incorporated

2. The submittal states that fittings are in accordance with ASTM D2467. Confirm that pressure fittings are in accordance with ASTM D2467, and DWV fittings will be in accordance with ASTM D2665 per Section 15230-2.02.A.2.b and c.

   Response - The vendor confirmed pressure fittings are in accordance with ASTM D2467. Also stated, "DWV" note is not applicable—DWV fittings are generally not used with SCH 80 pipe.

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   Incorporated - highlighted in "product marking" section

5. It is unclear which ASTM standard will be followed for solvent weld joint installation. Confirm that installation will be in accordance with ASTM D2855 per Section 15230-3.01.B.1.

   Incorporated - highlighted in "installation section" + complete instruction doc attached

SECTION 15230

PLASTIC PIPING AND TUBING

PART 1 GENERAL

1.01 SUMMARY

A. Section includes: Plastic pipe, tubing, and fittings.

1.02 REFERENCES

A. American Society of Mechanical Engineers (ASME):
   1. B16.12 - Cast Iron Threaded Drainage Fittings.

B. ASTM International (ASTM):
   1. D1248 - Standard Specification for Polyethylene Plastics Extrusion Materials for Wire and Cable.
   2. D1784 - Standard Specification for Rigid Poly(Vinyl Chloride) (PVC) Compounds and Chlorinated Poly(Vinyl Chloride) (CPVC) Compounds.
   3. D1785 - Standard Specification for Poly(Vinyl Chloride) (PVC) Plastic Pipe, Schedules 40, 80 and 120.
   4. D1869 - Standard Specification for Rubber Rings for Asbestos-Cement Pipe.
   5. D2241 - Standard Specification for Poly(Vinyl Chloride) (PVC) Pressure-Rated (SDR Series.
   6. D2412 - Standard Test Method for Determination of External Loading Characteristics of Plastic Pipe by Parallel-Plate Loading.
   7. D2466 - Standard Specification for Poly(Vinyl Chloride) (PVC) Plastic Pipe Fittings, Schedule 40.
   8. D2467 - Standard Specification for Poly(Vinyl Chloride) (PVC) Plastic Pipe Fittings, Schedule 80.
   9. D2513 - Standard Specification for Thermoplastic Gas Pressure Pipe, Tubing and Fittings.
   10. D2564 - Standard Specification for Solvent Cements for Poly(Vinyl Chloride) (PVC) Plastic Piping Systems.
   11. D2665 - Standard Specification for Poly(Vinyl Chloride) (PVC) Plastic Drain, Waste, and Vent Pipe and Fittings.
   12. D2855 - Standard Practice for Making Solvent-Cemented Joints with Poly (Vinyl Chloride)(PVC) Pipe and Fittings.
   13. D3034 - Standard Specification for Type PSM Poly(Vinyl Chloride) (PVC) Sewer Pipe and Fittings.
   14. D3212 - Standard Specification for Joints for Drain and Sewer Plastic Pipes Using Flexible Elastomeric Seals.
   15. D3261 - Standard Specification for Butt Heat Fusion Polyethylene (PE) Plastic Fittings for Polyethylene (PE) Plastic Pipe and Tubing.
   16. D3350 - Standard Specification for Polyethylene Plastic Pipes and Fittings Materials.
   17. D4101 - Standard Specification for Polypropylene Injection and Extrusion Materials.

18. F438 - Standard Specification for Socket-Type Chlorinated Poly(Vinyl Chloride) (CPVC) Plastic Pipe Fittings, Schedule 40.
19. F439 - Standard Specification for Chlorinated Poly(Vinyl Chloride) (CPVC) Plastic Pipe Fittings, Schedule 80.
20. F441 - Standard Specification for Chlorinated Poly(Vinyl Chloride) (CPVC) Plastic Pipe, Schedules 40 and 80.
21. F477 - Standard Specification for Elastomeric Seals (Gaskets) for Joining Plastic Pipe.
22. F493 - Standard Specification for Solvent Cements for Chlorinated Poly(Vinyl Chloride) (CPVC) Plastic Pipe and Fittings.
23. F645 - Standard Guide for Selection, Design and Installation of Thermoplastic Water-Pressure Piping Systems.
24. F679 - Standard Specification for Poly(Vinyl Chloride) (PVC) Large-Diameter Plastic Gravity Sewer Pipe and Fittings.
25. F714 - Standard Specification for Polyethylene (PE) Plastic Pipe (SDR-PR) Based on Outside Diameter.

C. American Water Works Association (AWWA):
   1. C900 - Standard for Polyvinyl Chloride (PVC) Pressure Pipe and Fabricated Fittings, 4 Inches to 12 Inches (100 mm Through 300 mm), for Water Transmission Distribution.

D. NSF International (NSF).

E. Plastics Pipe Institute (PPI):
   1. TR 31 - Underground Installation of Polyolefin Piping.

1.03 ABBREVIATIONS

A. ABS: Acrylonitrile-butadiene-styrene.

B. CPVC: Chlorinated polyvinyl chloride.

C. DR: Dimension ratio.

D. DWV: Drain, waste, and vent.

E. ID: Inside diameter of piping or tubing.

F. NPS: Nominal pipe size followed by the size designation.

G. NS: Nominal size of piping or tubing.

H. PE: Polyethylene.

I. PP: Polypropylene.

J. PVC: Polyvinyl chloride.

K. SDR: Standard dimension ratio; the outside diameter divided by the pipe wall thickness.

1.04 SUBMITTALS

A. Submit as specified in Section 01330 - Submittal Procedures.

B. Product data: As specified in Section 15052 - Common Work Results for General Piping.

C. Shop Drawings: will be submitted later
   1. Describe materials, pipe, fittings, gaskets, and solvent cement.
   2. Installation instructions.

1.05 QUALITY ASSURANCE

A. Plastic pipe in potable water applications: Provide pipe and tubing bearing NSF seal.

B. Fusion machine technician qualifications: 1-year experience in the installation of similar PE piping systems from the same manufacturer.

C. Mark plastic pipe with nominal size, type, class, schedule, or pressure rating, manufacturer and all markings required in accordance with ASTM and AWWA standards.

1.06 DELIVERY, STORAGE, AND HANDLING

A. Protect piping materials from sunlight, scoring, and distortion.

B. Do not allow surface temperatures on pipe and fittings to exceed 120 degrees Fahrenheit.

C. Store and handle PE pipe and fittings as recommended by manufacturer in published instructions.

PART 2 PRODUCTS

2.01 MATERIALS

A. Extruding and molding material: Virgin material containing no scrap, regrind, or rework material except where permitted in the referenced standards.

B. Fittings: Same material as the pipe and of equal or greater pressure rating, except that fittings used in drain, waste, and vent (DWV) piping systems need not be pressure rated:
   1. The definition of DWV piping systems on this project shall be limited to piping associated with toilets and sinks only.

C. Unions 2-1/2 inches and smaller: Socket end screwed unions. Make unions 3 inches and larger of socket flanges with 1/8-inch full-face soft EPDM gasket.

2.02 PVC PIPING, SCHEDULE TYPE

A. Materials:
   1. PVC Pipe: Designation PVC 1120 in accordance with ASTM D1785 and appendices:
      a. Pipe and fittings: Extruded from Type I, Grade 1, Class 12454-B material in accordance with ASTM D1784.
      b. PVC Pipe: Schedule 80 unless otherwise indicated on the Drawings or specified in the Piping Schedule in Section 15052 - Common Work Results for General Piping.
   2. Fittings:
      a. Supplied by pipe manufacturer.
      b. Pressure fittings: In accordance with ASTM D2466 or ASTM D2467.
      c. DWV fittings: In accordance with ASTM D2665.
   3. Solvent cement: In accordance with ASTM D2564:
      a. Manufacturers: The following or approved equal:
         1) IPS Corporation.
      b. Certified by the manufacturer for the service of the pipe.
      c. In potable water applications: Provide solvent cement listed by NSF 61 for potable water applications.
      d. Primer: As recommended by the solvent cement manufacturer.
      e. Chemical service: For CPVC or PVC pipe in chemical service, provide the following primer and cement, or approved equal:
         1) Primer: IPS Corp Type P70.
         2) Cement: IPS Corp Type 724 cement or another cement certified by the manufacturer for chemical service.
   4. Coating:
      a. Coat exposed piping as specified in Section 09960 - High-Performance Coatings.

2.03 PVC PIPING, CLASS TYPE

A. PVC pipe, Class Type: In accordance with AWWA C900:
   1. Pressure Class: 150 with a minimum DR of 18.
   2. Fittings: AWWA C900, DR18.
   3. Joint design: Push-on or mechanical joint type as identified in Piping Schedule.
   4. Gaskets: Neoprene in accordance with ASTM D1869 or ASTM F477.

2.04 CPVC PIPING

A. Materials:
   1. CPVC pipe: Schedule 40 or Schedule 80, as specified in the Piping Schedule in Section 15052 - Common Work Results for General Piping, in accordance with ASTM F441 and Appendix, CPVC 4120:
      a. Pipe: Extruded from Corzan Type IV, Grade 1, Class 24448 for sizes 8-inch and below and Class 23447 for sizes 8-inch and above material in accordance with ASTM D1784.
      b. Manufacturers: One of the following:
         1) IPEX USA, LLC.
         2) GF Harvel.

2. Fittings: In accordance with ASTM F438 or ASTM F439 for pressure fittings, as appropriate to the service and pressure requirement:
    a. Fittings: Supplied by the pipe manufacturer.
    b. Manufacturers: One of the following:
        1) IPEX USA, LLC.
        2) Chemtrol.
3. Solvent cement: In accordance with ASTM F493:
    a. Manufacturers: The following or approved equal:
        1) IPS Corporation.
    b. Certified by the manufacturer for the service of the pipe.
    c. Primer: As recommended by the solvent cement manufacturer.
    d. In potable water applications: Provide solvent cement listed by NSF 61 for potable water applications.
    e. For CPVC pipe in chemical service, provide the following primer and cement, or approved equal:
        1) Primer: IPS Corp Type P70.
        2) Cement: IPS Corp Type 724 cement or another cement certified by the manufacturer for high strength hypochlorite service.
    f. Contractor to submit solvent cement and primer selections prior to use.
4. Coating:
    a. Coat exposed piping as specified in Section 09960 - High-Performance Coatings.

2.05 PP PIPING

A. Materials:
1. Pipe: Schedule 40 dimensions, extruded from Type I-19509 material in accordance with ASTM D4101.
2. Fittings: Molded from the same material and same laying length in accordance with ASME B 16.12:
    a. Fittings: Manufactured by pipe manufacturer.
3. Coating:
    a. Coat exposed piping as specified in Section 09960 - High-Performance Coatings.

2.06 PE TUBING AND FITTINGS

A. Materials:
1. Small bore PE tubing: Black flexible virgin PE tubing, OD copper tubing size:
    a. Plastic tubing ID as follows:
        1) For NS 1/4 inch, ID of 0.170 inch.
        2) For NS 5/16 inch, ID of 0.187 inch.
        3) For NS 3/8 inch, ID of 0.251 inch.
        4) For NS of 1/2 inch, an ID of 0.375 inch.
2. Fittings: Compression fittings, Dekoron E-Z, or approved equal.
3. Protective sheath:
    a. Manufacturers: One of the following:
        1) Dekoron, "Poly-Cor."
        2) Parker Hannifin Corp./Fluid connector Products, Parflex® Division, Multitube.
4. Plug-in fittings for connection to instruments: Brass quick-connect fittings.

2.07 POLYETHYLENE PIPING FOR DRAIN, WASTE, AND VENT PIPING SYSTEMS

A. General:
   1. Pipe and fittings: High-density polyethylene.
   2. Dimensions of pipe and fittings: Based on controlled outside diameter in accordance with ASTM F714:
      a. SDR: Maximum of 11.

B. Manufacturers: One of the following:
   1. DuPont™, Sclairpipe®.
   2. Polaris, Dura-Tuff, Inc.

C. Pipe, fittings, and adapters: Furnished by the same manufacturer, and compatible with components in the same system and with components of other systems to which connected.

D. Materials:
   1. Polyethylene: In accordance with ASTM D1248, Type III, Class C, Category 5, Grade P34; listed by the Plastic Pipe Institute under the designation PE 3408; and have a minimum cell classification, in accordance with ASTM D3350.
   2. Pipe and fittings: Manufactured from material with the same cell classification.

E. Coating:
   1. Coat exposed piping as specified in Section 09960 - High-Performance Coatings.

2.08 SOURCE QUALITY CONTROL

A. PVC piping, Schedule Type:
   1. Mark pipe and fittings in accordance with ASTM D1785.

B. PVC piping, Class Type:
   1. Hydrostatic proof testing in accordance with AWWA C900: Test pipe and integral bell to withstand, without failure, two times the pressure class of the pipe for a minimum of 5 seconds.

C. CPVC piping:
   1. Mark pipe and fittings in accordance with ASTM F441.

D. PP piping:
   1. Test samples and testing: Cut test samples of pipe, 6 inches long, from full length sections and test by the method outlined in accordance with ASTM D2412:
      a. Deflect pipe at least 35 percent without failure. Stiffness at 5 percent deflection equals or exceeds 55 pounds per square inch after the test samples have been immersed in a 5 percent solution by weight of sulfuric acid and n-Heptane for a period of 24 hours prior to testing.
      b. Failure is defined as rupture of the pipe wall.
      c. Stiffness factor may be computed by the method outlined in accordance with ASTM D2412 or by dividing the load in pounds per linear inch by the deflection in inches and 5 percent deflection.

PART 3 EXECUTION

3.01 INSTALLATION

A. General:
1. Where not otherwise specified, install piping in accordance with ASTM F645, or manufacturer's published instructions for installation of piping, as applicable to the particular type of piping.
2. Provide molded transition fittings for transitions from plastic to metal or IPS pipe. Do not thread plastic pipe.
3. Locate unions where indicated on the Drawings, and elsewhere where required for adequate access and assembly of the piping system.
4. Provide serrated nipples for transition from plastic pipe to rubber hose.

B. Installation of PVC piping, Schedule Type:
1. Solvent weld joints in accordance with ASTM D2855:
   a. For PVC pipe in chemical service use IPS Corp. Type 724 cement and IPS Corp. Type P70 primer in accordance with manufacturer's instructions.
2. Install piping in accordance with manufacturer's published instructions.

C. Installation of PVC piping, Class Type:
1. Install piping in accordance with the Appendix of AWWA C900 complemented with manufacturer's published instructions.

D. Installation of CPVC piping:
1. Clean dirt and moisture from pipe and fittings.
2. Bevel pipe ends in accordance with manufacturer's instructions with chamfering tool or file. Remove burrs.
3. Use solvent cement and primer formulated for CPVC:
   a. For CPVC pipe in chemical service use IPS Corp. Type 724 cement and IPS Corp. Type P70 primer in accordance with manufacturer's instructions.
4. Use primer on pressure and non-pressure joints.
5. Do not solvent weld joints when ambient temperatures are below 40 degrees F or above 90 degrees F unless solvent cements specially formulated for these conditions are utilized.

E. Installation of PP piping:
1. Install piping in accordance with manufacturer's published instructions.

F. Installation of polyethylene (PE) tubing and fittings:
1. Install small bore PE tubing in accordance with manufacturer's printed instructions, in neat straight lines, supported at close enough intervals to avoid sagging, and in continuous runs wherever possible.
2. Bundle tubing in groups of parallel tubes within protective sheath.
3. Tubes within protective sheath may be color coded, but protect tubing other than black outside the sheath by wrapping with black plastic electrician's tape.
4. Grade tubing connected to meters in one direction.

G. Installation of PE piping for drain, waste, and vent:
    1. Install piping as recommended in manufacturer's published instructions.

3.02 FIELD QUALITY CONTROL

A. Test pipe as specified in Section 15052 - Common Work Results for General Piping and Section 15956 - Piping Systems Testing.

B. Leakage test for PVC piping, Class Type:
    1. Polyvinyl chloride (PVC) piping, Class Type: Subject to visible leaks test and to pressure test with maximum leakage allowance, as specified in Section 15956 - Piping Systems Testing.
    2. Pressure test with maximum leakage allowance: Perform test after backfilling:
        a. Pressure: 125 pounds per square inch, gauge.
        b. Maximum leakage allowance as follows, wherein the value for leakage is in gallons per 100 joints per hour:

NPS, Inches	**1-1/2**	**2**	**2-1/2**	**3**	**4**	**6**	**8**	**10**	**12**
Leakage	0.41	0.52	0.63	0.76	0.98	1.45	1.88	2.35	2.80

END OF SECTION

Submittal Information for Spears® Manufacturing Company
PVC Schedule 80 Solid Wall Pipe & Fitting System

Date: MAY 28, 2025 15230-2.02 A.1&2 **GSPVC80-0718**

Job Name: HILO WTTP **Location:** HILO, HI

Engineer: ______ **Contractor:** NAN, INC.

Scope:
This submittal covers Spears® PVC Schedule 80 solid wall pipe and fittings intended for use in pressure applications where the application operating temperature does not exceed 140° F (63°C).

Product Specification:
All Spears® PVC Schedule 80 fittings shall be manufactured in the U.S.A by Spears® Manufacturing Company from PVC Type I, with a minimum cell classification 12454 in accordance with ASTM Standard D1784. All injection molded PVC Schedule 80 fittings shall be certified for potable water service by NSF International manufactured in strict compliance to ASTM D2467. All fabricated fittings shall be produced in accordance with Spears® General Specifications for Fabricated Fittings (FAB-7). Spears® PVC Schedule 80 pipe and fittings shall be capable of withstanding a vacuum of twenty-six inches of mercury (Hg) at 73° F (23° C) when subjected to a one hour test with a leak factor of not more than one inch of Hg. All PVC flanges shall be designed and manufactured to meet CL150 bolt pattern per ANSI Standard B16.5 and rated for a maximum internal pressure of 150 psi, non-shock at 73°F unless otherwise noted.

All Spears® PVC Schedule 80 pipe shall be manufactured in the U.S.A by Spears® Manufacturing Company from a Type I, Grade I Polyvinyl Chloride (PVC) compound with a minimum cell classification of 12454 in accordance with ASTM D1784. The pipe shall be manufactured in strict compliance to ASTM D1785 consistently meeting and/or exceeding the quality assurance test requirements of these standards. All Spears® EverTUFF® pipe shall be manufactured in the USA and immediately wrapped for protection. The pipe shall be provided with plain ends in 20 foot cut lengths. All Spears® EverTUFF® pipe shall be certified by NSF International for potable water applications and marked accordingly.

Product Marking
All Spears® pipe shall be marked PVC schedule 80 and shall be marked with NSF® Listing, ASTM Standard and applicable pressure @ 73° F. (23°C). Spears® PVC Schedule 80 Fittings shall be engraved with markings required by ASTM Standard and bear an NSF® listing mark for potable water use.

Installation:
Installation for Spears® PVC Schedule 80 systems shall comply with current installation instructions published by Spears® Manufacturing Company, established industry practices and all applicable code requirements. Buried pipe shall be in accordance with ASTM 2774 and ASTM F1668. The piping system shall be joined using a two-step solvent cement joining process with primer conforming to ASTM F656 and solvent cement conforming to ASTM D2564. The system shall be protected from ultra violet (UV) light exposure from the sun or other source and protected from any chemicals that are not compatible with the PVC materials including but not limited to fire stopping materials, plasticizers, incompatible thread sealants etc.

NOTE: PVC piping systems are suitable for oil-free air handling to 25 psi, not for distribution of compressed air or gas.

Referenced Standards:
ASTM D1784 – Rigid Vinyl Compounds
ASTM D1785 – PVC Schedule 40, 80 & 120 Pipe
ASTM D2467 – PVC Schedule 80 Fittings
ASTM D2564 – Solvent Cements for PVC Pipe & Fittings
ASTM D2774 – Procedure for Buried Pressure Pipe
ASTM F656 – Primers for PVC Pipe & Fittings
ASTM F1668 – Procedures for Buried Plastic Pipe
ASTM F1866 – Fabricated PVC DWV Fittings
NSF International – Standard 14/61 Potable Water
ANSI B16.5 - Pipe Flange Dimensions

Approvals:
NSF® – NSF International Standard 14/61 Potable Water

Features:
Lightweight
Corrosion Resistant
Long Service Life

Spears® Technical Information

23-May-25 02:41:32 PM

Part No: PD-800-030 **Pipe** **Schedule 80 PVC & CPVC Pipe**

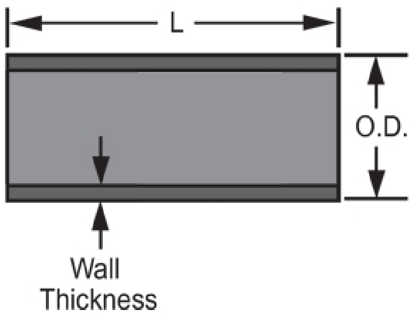
Desc: 3 PVC SCH80 PIPE 20 FT

MSRP: 22.61 **Part Code:** 744

Weight(lbs): 1.958 **Weight(kg):** 0.888 **Weight(gm):** 888

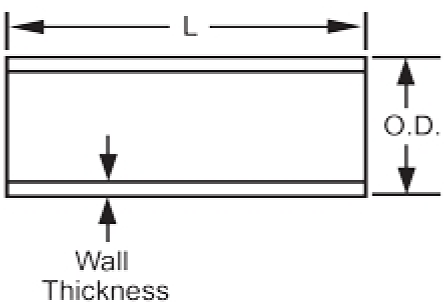
Size: 3" **Color:** GRAY **Material:** PVC

Pipe Type 20 Foot Length



L	= 240	**AVG.O.D.**	= 3.500	**Minimum Wall**	= .300

Typical Pipe Dimension Reference



Spears® Technical Information

19-Jul-22 08:59:12 AM

Part No: 801-030 **Schedule 80 PVC & CPVC** **Tee**

Desc: 3 PVC TEE SOC SCH80

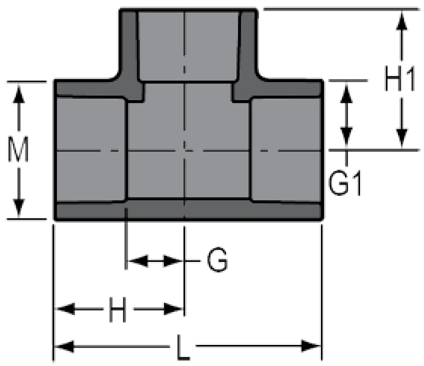
MSRP: 119.2 **Part Code:** 080

Weight(lbs): 2.147 **Weight(kg):** 0.974 **Weight(gm):** 974

Size: 3" **Color:** GRAY **Material:** PVC

Connection Socket x Socket x Socket

Type Standard



G = 2- 1/16 **G1** = 2- 1/16 **H** = 3-31/32 **H1** = 3-31/32 **L** = 7-15/16

M = 4- 5/32

Injection Molded Dimension References:

G = (LAYING LENGTH) intersection of center lines to bottom of socket/thread; 90° elbows, tees, crosses; ± 1/32 inch.

H = Intersection of center lines to face of fitting; 90° elbows tees, crosses; ± 1/32 inch.

J = Intersection of center lines to bottom of socket/thread; 45° elbows; ± 1/32 inch

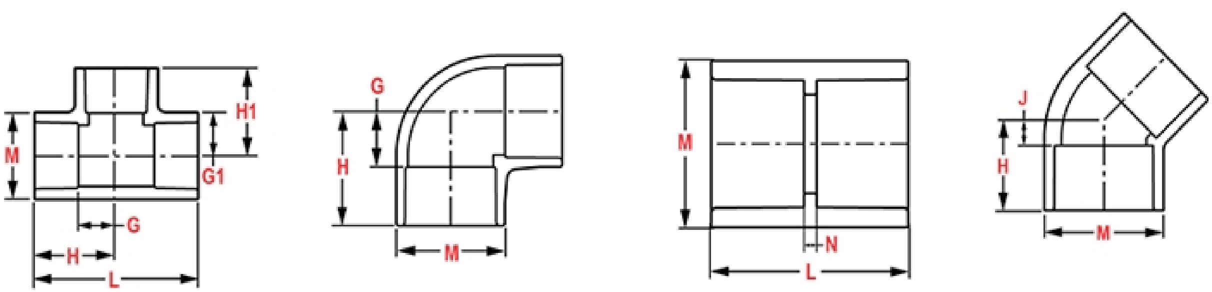
L = Overall length of fittings; ± 1/16 inch.

M = Outside diameter of socket/thread hub; ± 1/16 inch.

N = Socket bottom to socket bottom; couplings; ± 1/16 inch.

W = Height of cap; ± 1/16 inch

Typical Molded Dimension References



Spears® Technical Information

23-May-25 03:17:11 PM

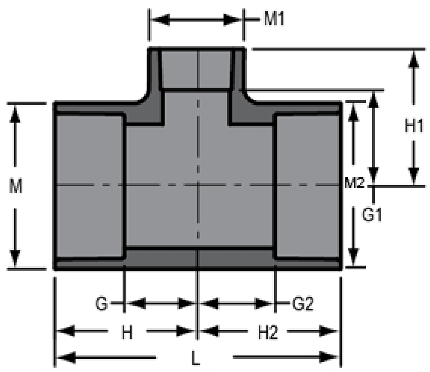
Part No: 801-337 **Schedule 80 PVC & CPVC** **Tee**

Desc: 3X1-1/2 PVC RED TEE SOC SCH80

MSRP: 114.49 **Part Code:** 080

Weight(lbs): 1.442 **Weight(kg):** 0.654 **Weight(gm):** 654

Size: 3"X3"X1-1/2" **Color:** GRAY **Material:** PVC



Connection Socket x Socket x Socket

Type Reduction

G = 1- 5/32	**G1** = 2- 1/16	**G2** = 1- 5/32	**M2** = 4-3/16	**H** = 3- 1/16
H1 = 3- 7/16	**H2** = 3- 1/16	**L** = 6-1/8	**M** = 4- 3/16	**M1** = 2- 3/8

Injection Molded Dimension References:

G = (LAYING LENGTH) intersection of center lines to bottom of socket/thread; 90° elbows, tees, crosses; ± 1/32 inch.

H = Intersection of center lines to face of fitting; 90° elbows tees, crosses; ± 1/32 inch.

J = Intersection of center lines to bottom of socket/thread; 45° elbows; ± 1/32 inch

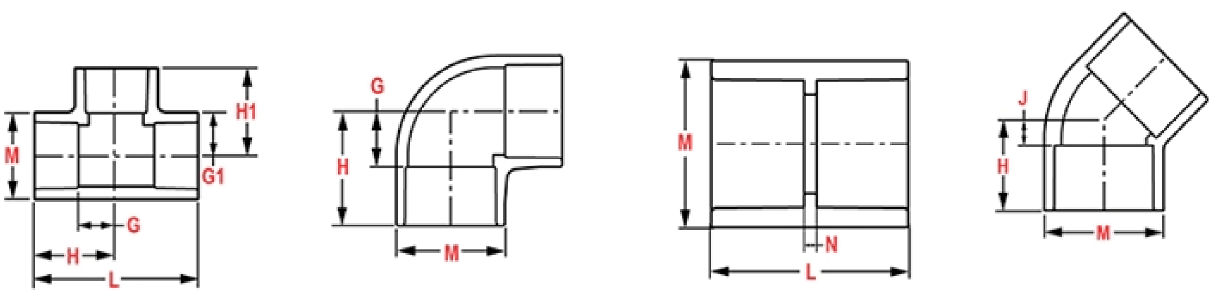
L = Overall length of fittings; ± 1/16 inch.

M = Outside diameter of socket/thread hub; ± 1/16 inch.

N = Socket bottom to socket bottom; couplings; ± 1/16 inch.

W = Height of cap; ± 1/16 inch

Typical Molded Dimension References



The information printed here is based on current information & product design at the time of publication and is subject to change without notification. Spears® ongoing commitment to product improvement may result in some variation. No representation, guarantees or warranties of any kind are as to its accuracy, suitability for particular application or results to be obtained thereform. For verification of technical data or additional information, please contact Spears® Technical Service Department :: WEST COAST : (818) 364-1611 - EAST COAST : (717) 938-9006

Spears® Technical Information

19-Jul-22 09:09:30 AM

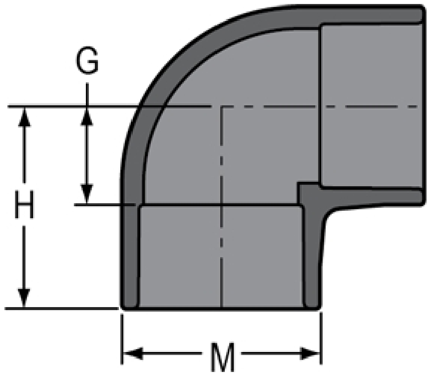
Part No: 806-030 **Schedule 80 PVC & CPVC** **Elbow**

Desc: 3 PVC 90 ELL SOC SCH80

MSRP: 64.8 **Part Code:** 080

Weight(lbs): 1.671 **Weight(kg):** 0.758 **Weight(gm):** 758

Size: 3" **Color:** GRAY **Material:** PVC



Connection	Socket x Socket
Angle	90°
Type	Standard

G = 2- 1/16 **H** = 4 **M** = 4- 3/16

Injection Molded Dimension References:

G = (LAYING LENGTH) intersection of center lines to bottom of socket/thread; 90° elbows, tees, crosses; ± 1/32 inch.

H = Intersection of center lines to face of fitting; 90° elbows tees, crosses; ± 1/32 inch.

J = Intersection of center lines to bottom of socket/thread; 45° elbows; ± 1/32 inch

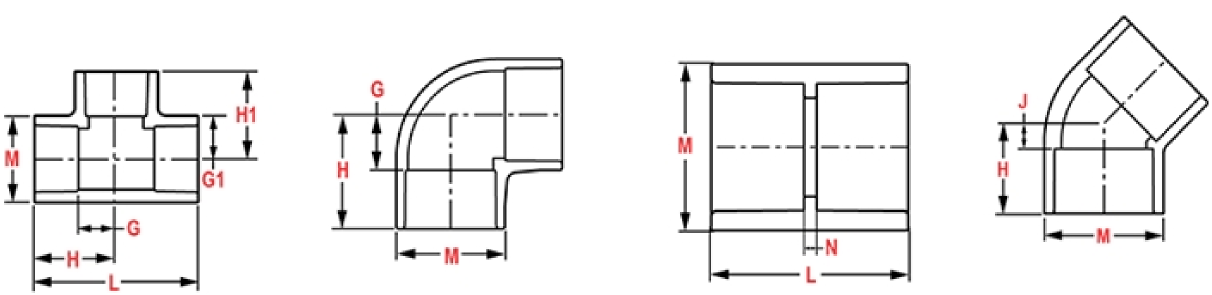
L = Overall length of fittings; ± 1/16 inch.

M = Outside diameter of socket/thread hub; ± 1/16 inch.

N = Socket bottom to socket bottom; couplings; ± 1/16 inch.

W = Height of cap; ± 1/16 inch

Typical Molded Dimension References



Spears® Technical Information

23-May-25 02:55:05 PM

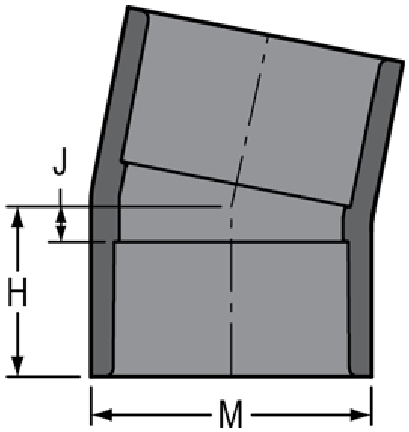
Part No: 811-030 **Schedule 80 PVC & CPVC** **Elbow**

Desc: 3 PVC 11-1/4 ELL SOC SCH80

MSRP: 184.41 **Part Code:** 080

Weight(lbs): 0.877 **Weight(kg):** 0.398 **Weight(gm):** 398

Size: 3" **Color:** GRAY **Material:** PVC



Connection	Socket x Socket
Angle	11-1/4°
Type	Standard

H = 2- 3/16 **J** = 5/16 **M** = 4-5/32

Injection Molded Dimension References:

G = (LAYING LENGTH) intersection of center lines to bottom of socket/thread; 90° elbows, tees, crosses; ± 1/32 inch.

H = Intersection of center lines to face of fitting; 90° elbows tees, crosses; ± 1/32 inch.

J = Intersection of center lines to bottom of socket/thread; 45° elbows; ± 1/32 inch

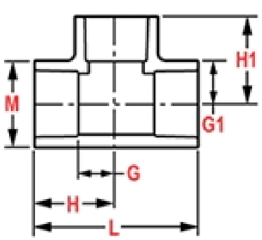
L = Overall length of fittings; ± 1/16 inch.

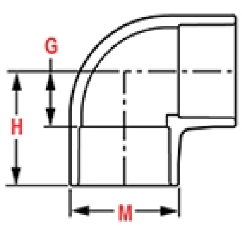
M = Outside diameter of socket/thread hub; ± 1/16 inch.

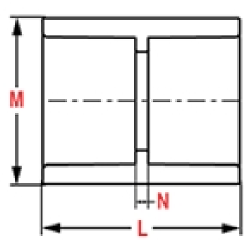
N = Socket bottom to socket bottom; couplings; ± 1/16 inch.

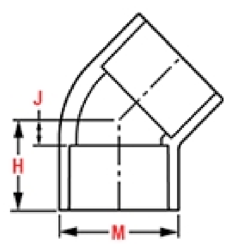
W = Height of cap; ± 1/16 inch

Typical Molded Dimension References









Spears® Technical Information

23-May-25 02:54:04 PM

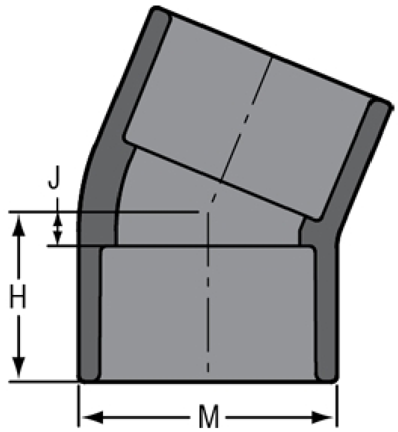
Part No: 816-030 **Schedule 80 PVC & CPVC** **Elbow**

Desc: 3 PVC 22-1/2 ELL SOC SCH80

MSRP: 184.41 **Part Code:** 080

Weight(lbs): 0.968 **Weight(kg):** 0.439 **Weight(gm):** 439

Size: 3" **Color:** GRAY **Material:** PVC



Connection	Socket x Socket
Angle	22-1/2°
Type	Standard

H = 2-3/8 **M** = 4- 5/32 **J** = 1/2

Injection Molded Dimension References:

G = (LAYING LENGTH) intersection of center lines to bottom of socket/thread; 90° elbows, tees, crosses; ± 1/32 inch.

H = Intersection of center lines to face of fitting; 90° elbows tees, crosses; ± 1/32 inch.

J = Intersection of center lines to bottom of socket/thread; 45° elbows; ± 1/32 inch

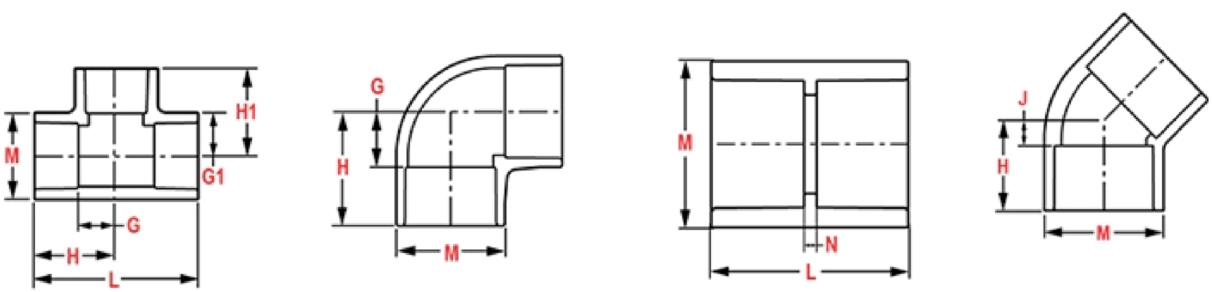
L = Overall length of fittings; ± 1/16 inch.

M = Outside diameter of socket/thread hub; ± 1/16 inch.

N = Socket bottom to socket bottom; couplings; ± 1/16 inch.

W = Height of cap; ± 1/16 inch

Typical Molded Dimension References



Spears® Technical Information

23-May-25 02:53:06 PM

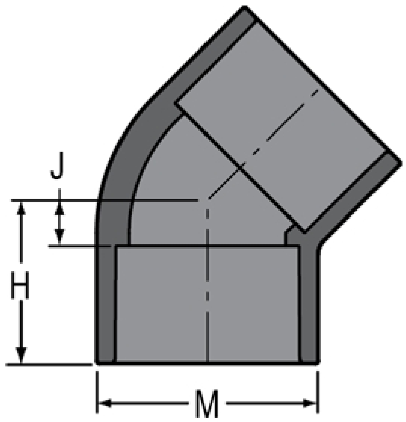
Part No: 817-030 **Schedule 80 PVC & CPVC** **Elbow**

Desc: 3 PVC 45 ELL SOC SCH80

MSRP: 156.44 **Part Code:** 080

Weight(lbs): 1.116 **Weight(kg):** 0.506 **Weight(gm):** 506

Size: 3" **Color:** GRAY **Material:** PVC



Connection	Socket x Socket
Angle	45°
Type	Standard

H = 2-25/32 **J** = 29/32 **M** = 4-1/8

Injection Molded Dimension References:

G = (LAYING LENGTH) intersection of center lines to bottom of socket/thread; 90° elbows, tees, crosses; ± 1/32 inch.

H = Intersection of center lines to face of fitting; 90° elbows tees, crosses; ± 1/32 inch.

J = Intersection of center lines to bottom of socket/thread; 45° elbows; ± 1/32 inch

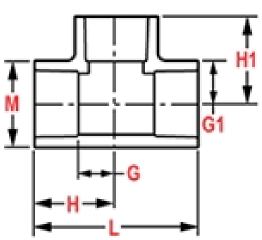
L = Overall length of fittings; ± 1/16 inch.

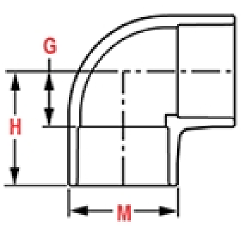
M = Outside diameter of socket/thread hub; ± 1/16 inch.

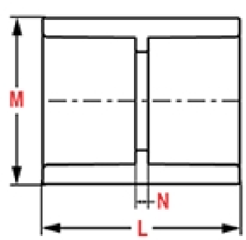
N = Socket bottom to socket bottom; couplings; ± 1/16 inch.

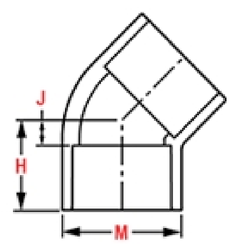
W = Height of cap; ± 1/16 inch

Typical Molded Dimension References









Spears® Technical Information

23-May-25 03:19:16 PM

Part No: 829-337 **Schedule 80 PVC & CPVC** **Coupling**

Desc: 3X1-1/2 PVC RED COUPLING SOC SCH80

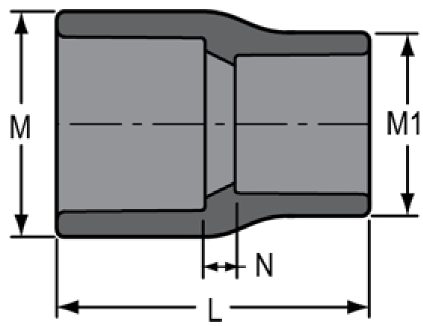
MSRP: 171.61 **Part Code:** 080

Weight(lbs): 0.692 **Weight(kg):** 0.314 **Weight(gm):** 314

Size: 3"X1-1/2" **Color:** GRAY **Material:** PVC

Connection Socket x Socket

Type Reduction



M = 4- 3/16 **M1** = 2- 3/8 **N** = 7/32 **L** = 3-1/2

Injection Molded Dimension References:

G = (LAYING LENGTH) intersection of center lines to bottom of socket/thread; 90° elbows, tees, crosses; ± 1/32 inch.

H = Intersection of center lines to face of fitting; 90° elbows tees, crosses; ± 1/32 inch.

J = Intersection of center lines to bottom of socket/thread; 45° elbows; ± 1/32 inch

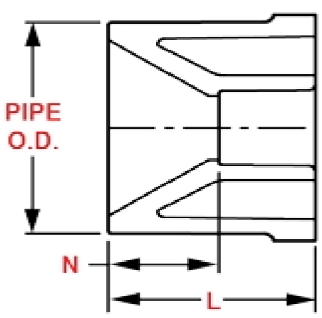
L = Overall length of fittings; ± 1/16 inch.

M = Outside diameter of socket/thread hub; ± 1/16 inch.

N = Socket bottom to socket bottom; couplings; ± 1/16 inch.

W = Height of cap; ± 1/16 inch

Typical Bushing Dimension References



Spears® Technical Information

28-May-25 02:38:59 PM

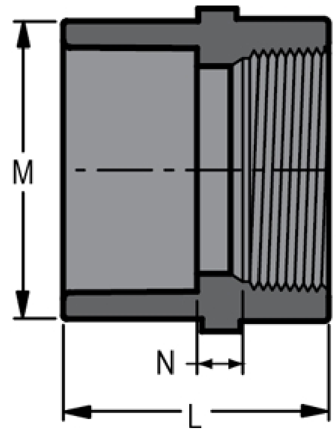
Part No: 835-030 **Schedule 80 PVC & CPVC** **Adapter**

Desc: 3 PVC FEMALE ADAPTER SOCXFPT SCH80

MSRP: 166.07 **Part Code:** 080

Weight(lbs): 0.747 **Weight(kg):** 0.339 **Weight(gm):** 339

Size: 3" **Color:** GRAY **Material:** PVC



Connection Socket x Fipt

Type Standard

L = 3-1/2 **M** = 4-5/32 **N** = 7/32

Injection Molded Dimension References:

G = (LAYING LENGTH) intersection of center lines to bottom of socket/thread; 90° elbows, tees, crosses; ± 1/32 inch.

H = Intersection of center lines to face of fitting; 90° elbows tees, crosses; ± 1/32 inch.

J = Intersection of center lines to bottom of socket/thread; 45° elbows; ± 1/32 inch

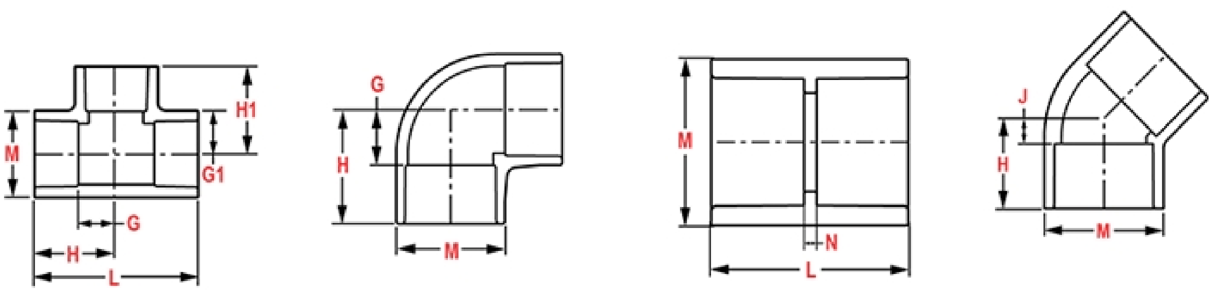
L = Overall length of fittings; ± 1/16 inch.

M = Outside diameter of socket/thread hub; ± 1/16 inch.

N = Socket bottom to socket bottom; couplings; ± 1/16 inch.

W = Height of cap; ± 1/16 inch

Typical Molded Dimension References



Spears® Technical Information

23-May-25 03:12:05 PM

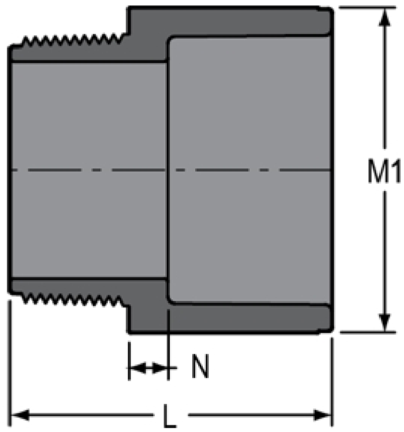
Part No: 836-030 **Schedule 80 PVC & CPVC** **Adapter**

Desc: 3 PVC MALE ADAPTER MPTXSOC SCH80

MSRP: 89.93 **Part Code:** 080

Weight(lbs): 0.745 **Weight(kg):** 0.338 **Weight(gm):** 338

Size: 3" **Color:** GRAY **Material:** PVC



Connection	Mipt x Socket
Type	Standard

L = 3-29/32 **M1** = 4-7/32 **N** = 3/8

Injection Molded Dimension References:

G = (LAYING LENGTH) intersection of center lines to bottom of socket/thread; 90° elbows, tees, crosses; ± 1/32 inch.

H = Intersection of center lines to face of fitting; 90° elbows tees, crosses; ± 1/32 inch.

J = Intersection of center lines to bottom of socket/thread; 45° elbows; ± 1/32 inch

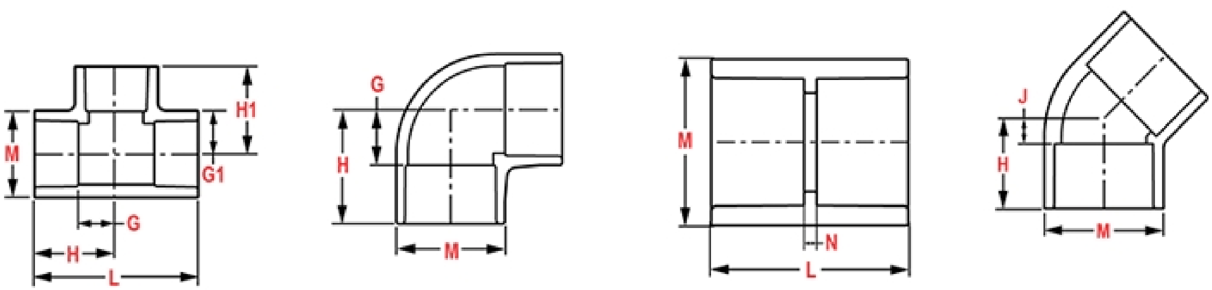
L = Overall length of fittings; ± 1/16 inch.

M = Outside diameter of socket/thread hub; ± 1/16 inch.

N = Socket bottom to socket bottom; couplings; ± 1/16 inch.

W = Height of cap; ± 1/16 inch

Typical Molded Dimension References



Spears® Technical Information

28-May-25 02:33:29 PM

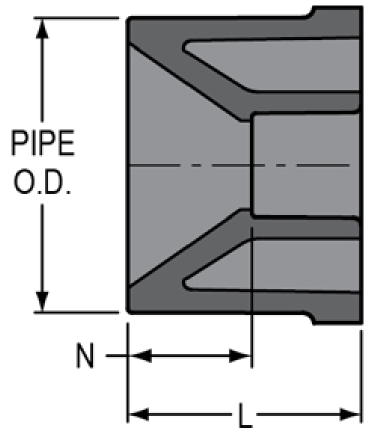
Part No: 837-335 **Schedule 80 PVC & CPVC** **Bushing**

Desc: 3X1 PVC RED BUSHING SPIGOTXSOC SCH80

MSRP: 101.87 **Part Code:** 080

Weight(lbs): 0.591 **Weight(kg):** 0.268 **Weight(gm):** 268

Size: 3"X1" **Color:** GRAY **Material:** PVC



Connection Spigot x Socket

Type Reduction

L = 2-3/16 **N** = 1-1/16

Injection Molded Dimension References:

G = (LAYING LENGTH) intersection of center lines to bottom of socket/thread; 90° elbows, tees, crosses; ± 1/32 inch.

H = Intersection of center lines to face of fitting; 90° elbows tees, crosses; ± 1/32 inch.

J = Intersection of center lines to bottom of socket/thread; 45° elbows; ± 1/32 inch

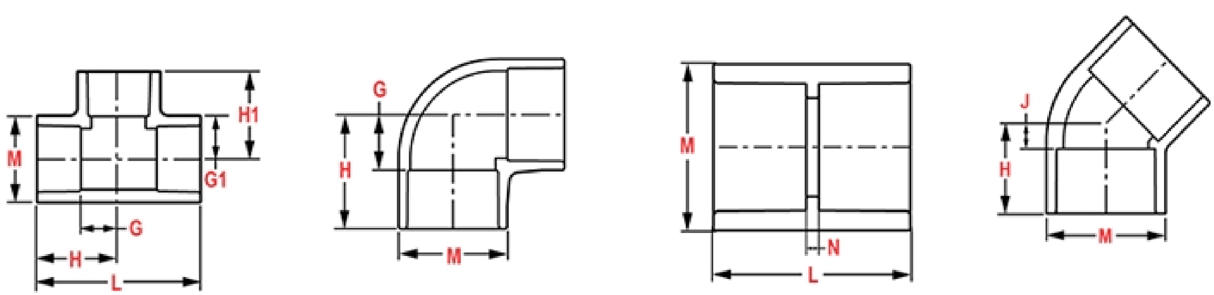
L = Overall length of fittings; ± 1/16 inch.

M = Outside diameter of socket/thread hub; ± 1/16 inch.

N = Socket bottom to socket bottom; couplings; ± 1/16 inch.

W = Height of cap; ± 1/16 inch

Typical Molded Dimension References



Spears® Technical Information

23-May-25 02:40:15 PM

Part No: PD-800-020 **Pipe** **Schedule 80 PVC & CPVC Pipe**

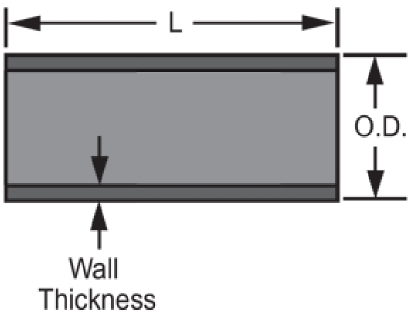
Desc: 2 PVC SCH80 PIPE 20 FT

MSRP: 11.04 **Part Code:** 744

Weight(lbs): 0.985 **Weight(kg):** 0.447 **Weight(gm):** 447

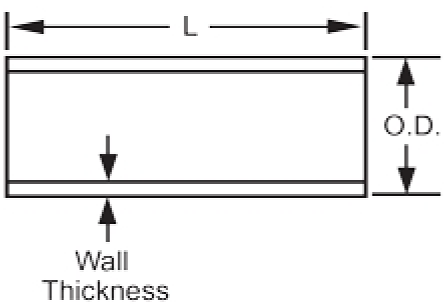
Size: 2" **Color:** GRAY **Material:** PVC

Pipe Type 20 Foot Length



L = 240	**AVG.O.D.** = 2.375	**Minimum Wall** = .218

Typical Pipe Dimension Reference



Spears® Technical Information

19-Jul-22 09:41:29 AM

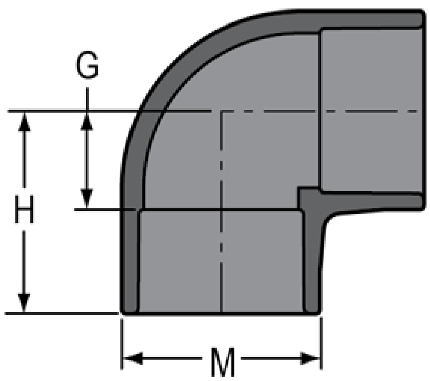
Part No: 806-020 **Schedule 80 PVC & CPVC** **Elbow**

Desc: 2 PVC 90 ELL SOC SCH80

MSRP: 24.62 **Part Code:** 080

Weight(lbs): 0.606 **Weight(kg):** 0.275 **Weight(gm):** 275

Size: 2" **Color:** GRAY **Material:** PVC



Connection	Socket x Socket
Angle	90°
Type	Standard

G = 1-13/32 **H** = 2-15/16 **M** = 2- 7/8

Injection Molded Dimension References:

G = (LAYING LENGTH) intersection of center lines to bottom of socket/thread; 90° elbows, tees, crosses; ± 1/32 inch.

H = Intersection of center lines to face of fitting; 90° elbows tees, crosses; ± 1/32 inch.

J = Intersection of center lines to bottom of socket/thread; 45° elbows; ± 1/32 inch

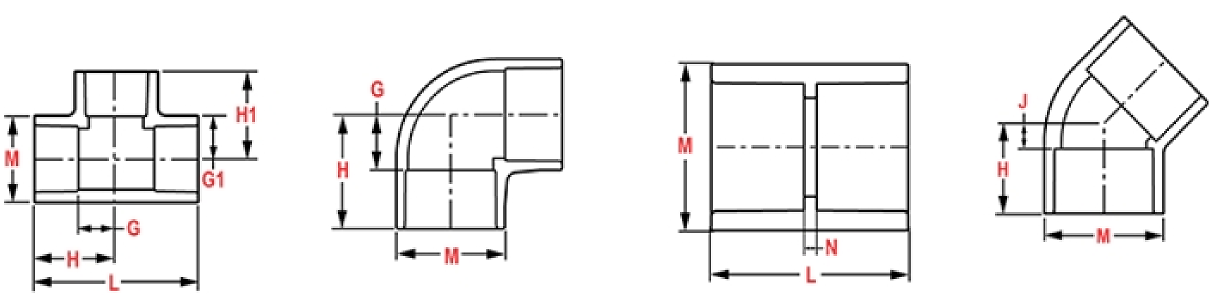
L = Overall length of fittings; ± 1/16 inch.

M = Outside diameter of socket/thread hub; ± 1/16 inch.

N = Socket bottom to socket bottom; couplings; ± 1/16 inch.

W = Height of cap; ± 1/16 inch

Typical Molded Dimension References



Spears® Technical Information

23-May-25 02:50:55 PM

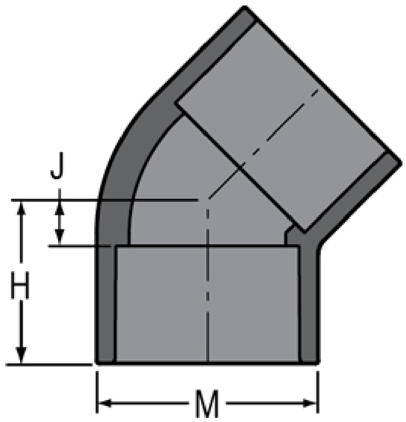
Part No: 817-020 **Schedule 80 PVC & CPVC** **Elbow**

Desc: 2 PVC 45 ELL SOC SCH80

MSRP: 61.2 **Part Code:** 080

Weight(lbs): 0.456 **Weight(kg):** 0.207 **Weight(gm):** 207

Size: 2" **Color:** GRAY **Material:** PVC



Connection	Socket x Socket
Angle	45°
Type	Standard

H = 2- 1/8 **J** = 5/8 **M** = 2- 7/8

Injection Molded Dimension References:

G = (LAYING LENGTH) intersection of center lines to bottom of socket/thread; 90° elbows, tees, crosses; ± 1/32 inch.

H = Intersection of center lines to face of fitting; 90° elbows tees, crosses; ± 1/32 inch.

J = Intersection of center lines to bottom of socket/thread; 45° elbows; ± 1/32 inch

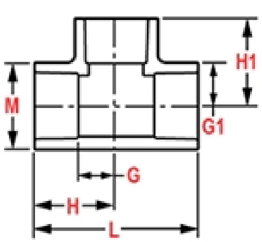
L = Overall length of fittings; ± 1/16 inch.

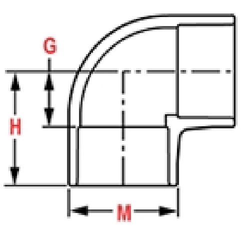
M = Outside diameter of socket/thread hub; ± 1/16 inch.

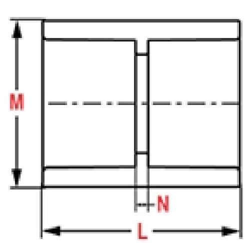
N = Socket bottom to socket bottom; couplings; ± 1/16 inch.

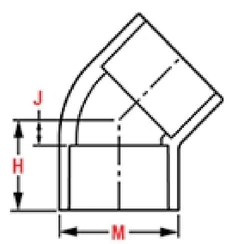
W = Height of cap; ± 1/16 inch

Typical Molded Dimension References









Spears® Technical Information

23-Aug-23 10:41:05 PM

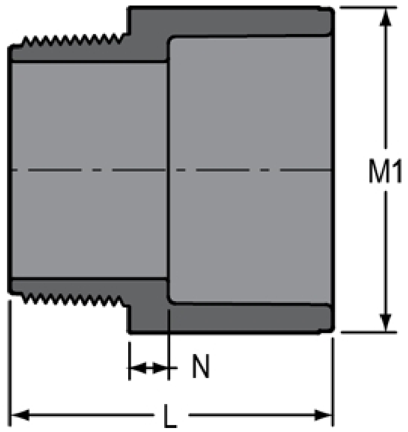
Part No: 836-020 **Schedule 80 PVC & CPVC** **Adapter**

Desc: 2 PVC MALE ADAPTER MPTXSOC SCH80

MSRP: 67.72 **Part Code:** 080

Weight(lbs): 0.282 **Weight(kg):** 0.128 **Weight(gm):** 128

Size: 2" **Color:** GRAY **Material:** PVC



Connection Mipt x Socket

Type Standard

L = 2- 7/8 **M1** = 2- 7/8 **N** = 5/16

Injection Molded Dimension References:

G = (LAYING LENGTH) intersection of center lines to bottom of socket/thread; 90° elbows, tees, crosses; ± 1/32 inch.

H = Intersection of center lines to face of fitting; 90° elbows tees, crosses; ± 1/32 inch.

J = Intersection of center lines to bottom of socket/thread; 45° elbows; ± 1/32 inch

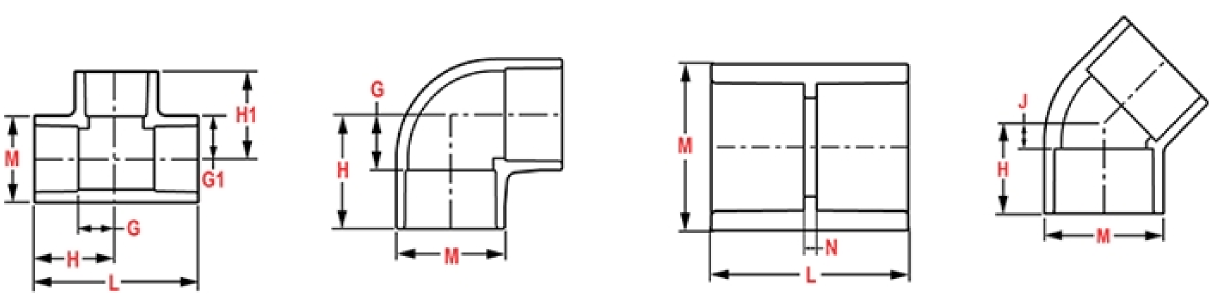
L = Overall length of fittings; ± 1/16 inch.

M = Outside diameter of socket/thread hub; ± 1/16 inch.

N = Socket bottom to socket bottom; couplings; ± 1/16 inch.

W = Height of cap; ± 1/16 inch

Typical Molded Dimension References



Spears® Technical Information

23-May-25 02:42:52 PM

Part No: PD-800-015 **Pipe** **Schedule 80 PVC & CPVC Pipe**

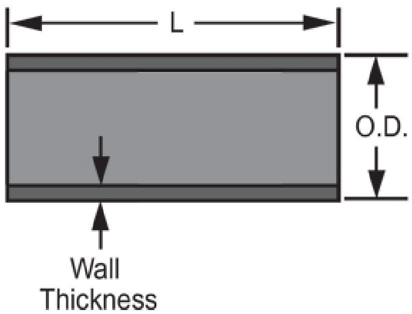
Desc: 1-1/2 PVC SCH80 PIPE 20 FT

MSRP: 7.98 **Part Code:** 744

Weight(lbs): 0.701 **Weight(kg):** 0.318 **Weight(gm):** 318

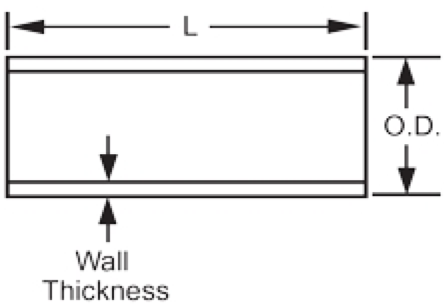
Size: 1-1/2" **Color:** GRAY **Material:** PVC

Pipe Type 20 Foot Length



AVG.O.D.	=	1.900	**Minimum Wall**	=	.200	**L**	= 240

Typical Pipe Dimension Reference



Spears® Technical Information

05-Aug-22 02:46:37 PM

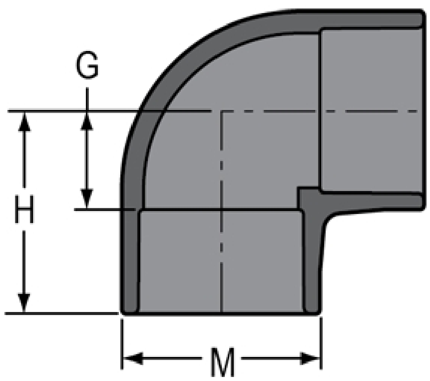
Part No: 806-015 **Schedule 80 PVC & CPVC** **Elbow**

Desc: 1-1/2 PVC 90 ELL SOC SCH80

MSRP: 20.38 **Part Code:** 080

Weight(lbs): 0.386 **Weight(kg):** 0.175 **Weight(gm):** 175

Size: 1-1/2" **Color:** GRAY **Material:** PVC



Connection	Socket x Socket
Angle	90°
Type	Standard

G = 1- 5/32 **H** = 2- 9/16 **M** = 2-11/32

Injection Molded Dimension References:

G = (LAYING LENGTH) intersection of center lines to bottom of socket/thread; 90° elbows, tees, crosses; ± 1/32 inch.

H = Intersection of center lines to face of fitting; 90° elbows tees, crosses; ± 1/32 inch.

J = Intersection of center lines to bottom of socket/thread; 45° elbows; ± 1/32 inch

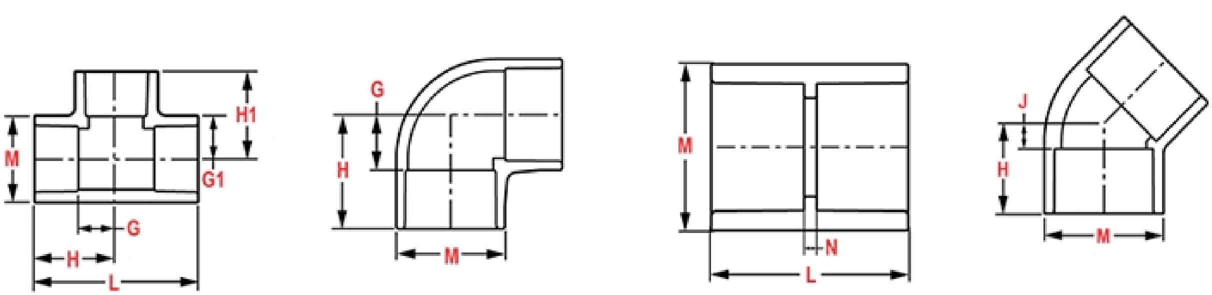
L = Overall length of fittings; ± 1/16 inch.

M = Outside diameter of socket/thread hub; ± 1/16 inch.

N = Socket bottom to socket bottom; couplings; ± 1/16 inch.

W = Height of cap; ± 1/16 inch

Typical Molded Dimension References



Spears® Technical Information

23-May-25 03:10:52 PM

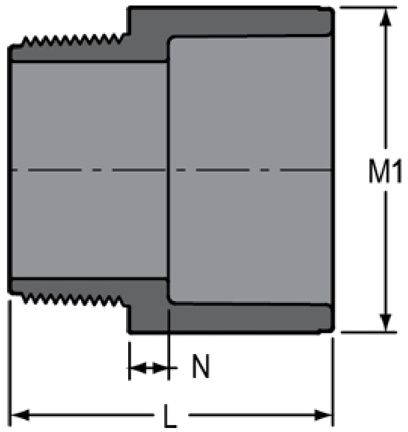
Part No: 836-015 **Schedule 80 PVC & CPVC** **Adapter**

Desc: 1-1/2 PVC MALE ADAPTER MPTXSOC SCH80

MSRP: 49.29 **Part Code:** 080

Weight(lbs): 0.19 **Weight(kg):** 0.086 **Weight(gm):** 86

Size: 1-1/2" **Color:** GRAY **Material:** PVC



Connection	Mipt x Socket
Type	Standard

L = 2-11/16 **M1** = 2- 3/8 **N** = 9/32

Injection Molded Dimension References:

G = (LAYING LENGTH) intersection of center lines to bottom of socket/thread; 90° elbows, tees, crosses; ± 1/32 inch.

H = Intersection of center lines to face of fitting; 90° elbows tees, crosses; ± 1/32 inch.

J = Intersection of center lines to bottom of socket/thread; 45° elbows; ± 1/32 inch

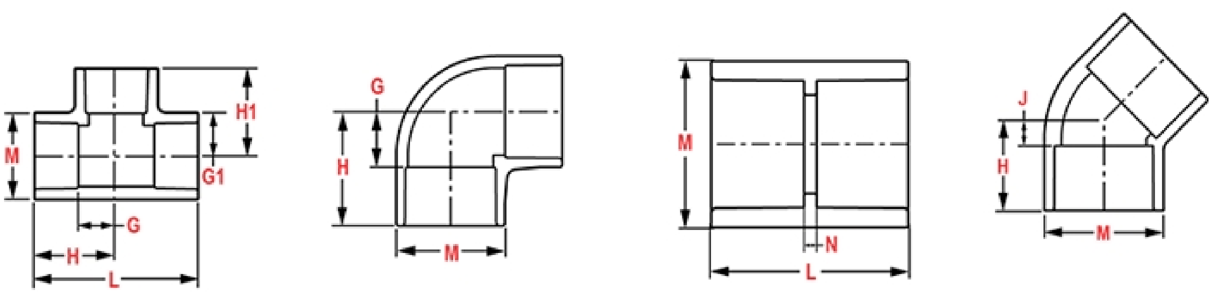
L = Overall length of fittings; ± 1/16 inch.

M = Outside diameter of socket/thread hub; ± 1/16 inch.

N = Socket bottom to socket bottom; couplings; ± 1/16 inch.

W = Height of cap; ± 1/16 inch

Typical Molded Dimension References



Spears® Technical Information

23-May-25 03:08:13 PM

Part No: 847-015 **Schedule 80 PVC & CPVC** **Cap/Plug**

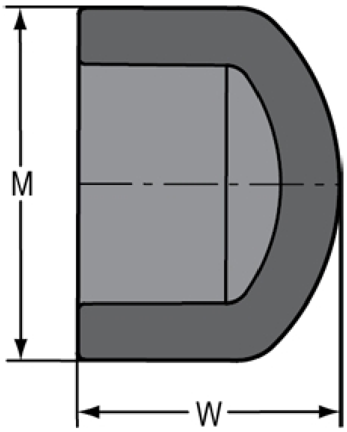
Desc: 1-1/2 PVC CAP SOC SCH80

MSRP: 29.07 **Part Code:** 080

Weight(lbs): 0.163 **Weight(kg):** 0.074 **Weight(gm):** 74

Size: 1-1/2" **Color:** GRAY **Material:** PVC

Connection Socket (Cap)



M = 2-11/32 **W** = 2

Injection Molded Dimension References:

G = (LAYING LENGTH) intersection of center lines to bottom of socket/thread; 90° elbows, tees, crosses; ± 1/32 inch.

H = Intersection of center lines to face of fitting; 90° elbows tees, crosses; ± 1/32 inch.

J = Intersection of center lines to bottom of socket/thread; 45° elbows; ± 1/32 inch

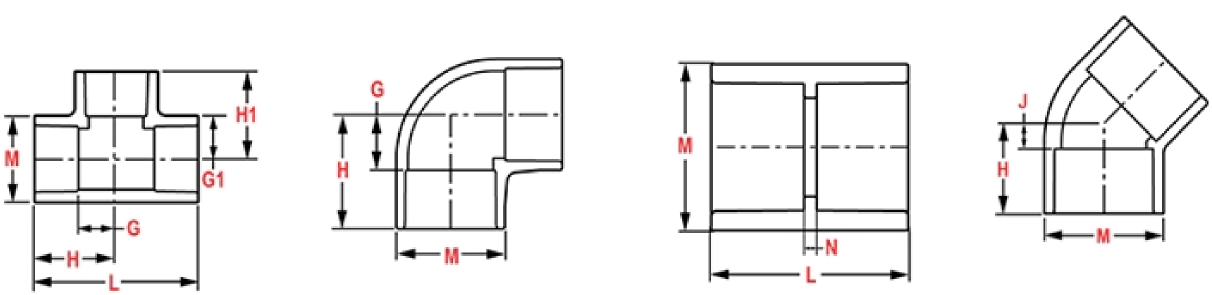
L = Overall length of fittings; ± 1/16 inch.

M = Outside diameter of socket/thread hub; ± 1/16 inch.

N = Socket bottom to socket bottom; couplings; ± 1/16 inch.

W = Height of cap; ± 1/16 inch

Typical Molded Dimension References



Spears® Technical Information

23-May-25 02:46:22 PM

Part No: PD-800-010 **Pipe** **Schedule 80 PVC & CPVC Pipe**

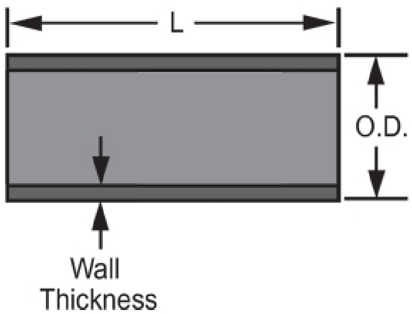
Desc: 1 PVC SCH80 PIPE 20 FT

MSRP: 5.12 **Part Code:** 744

Weight(lbs): 0.408 **Weight(kg):** 0.185 **Weight(gm):** 185

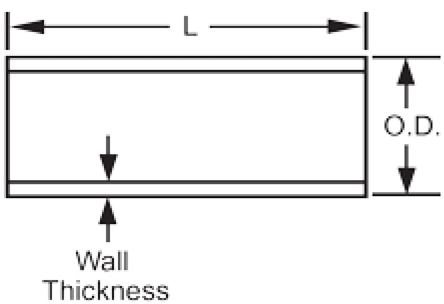
Size: 1" **Color:** GRAY **Material:** PVC

Pipe Type 20 Foot Length



L = 240 **AVG.O.D.** = 1.315 **Minimum Wall** = .179

Typical Pipe Dimension Reference



Spears® Technical Information

28-May-25 02:42:32 PM

Part No: 801-010 **Schedule 80 PVC & CPVC** **Tee**

Desc: 1 PVC TEE SOC SCH80

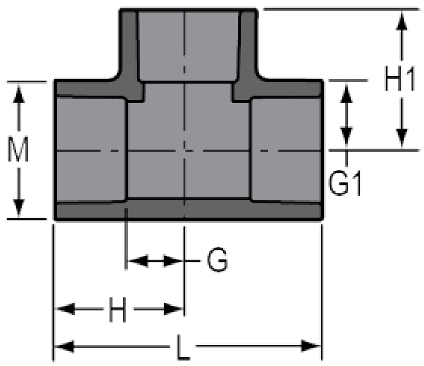
MSRP: 26.87 **Part Code:** 080

Weight(lbs): 0.278 **Weight(kg):** 0.126 **Weight(gm):** 126

Size: 1" **Color:** GRAY **Material:** PVC

Connection Socket x Socket x Socket

Type Standard



G = 7/8 **G1** = 7/8 **H** = 2 **H1** = 2 **L** = 4

M = 1-23/32

Injection Molded Dimension References:

G = (LAYING LENGTH) intersection of center lines to bottom of socket/thread; 90° elbows, tees, crosses; ± 1/32 inch.

H = Intersection of center lines to face of fitting; 90° elbows tees, crosses; ± 1/32 inch.

J = Intersection of center lines to bottom of socket/thread; 45° elbows; ± 1/32 inch

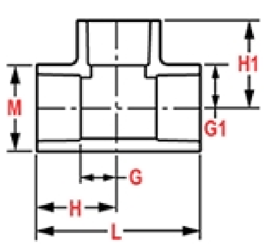
L = Overall length of fittings; ± 1/16 inch.

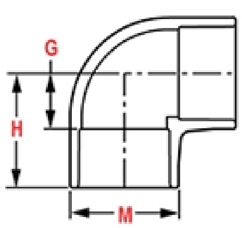
M = Outside diameter of socket/thread hub; ± 1/16 inch.

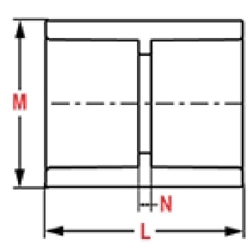
N = Socket bottom to socket bottom; couplings; ± 1/16 inch.

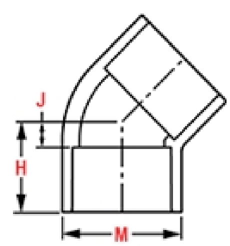
W = Height of cap; ± 1/16 inch

Typical Molded Dimension References









Spears® Technical Information

10-Sep-24 02:11:00 PM

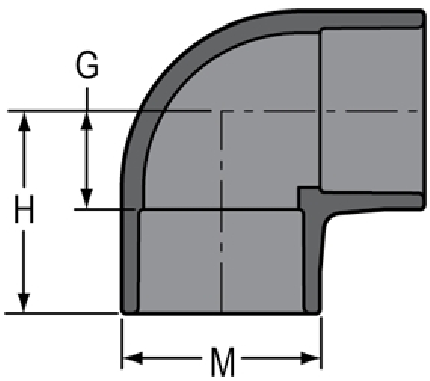
Part No: 806-010 **Schedule 80 PVC & CPVC** **Elbow**

Desc: 1 PVC 90 ELL SOC SCH80

MSRP: 14.97 **Part Code:** 080

Weight(lbs): 0.192 **Weight(kg):** 0.087 **Weight(gm):** 87

Size: 1" **Color:** GRAY **Material:** PVC



Connection	Socket x Socket
Angle	90°
Type	Standard

G = 7/8 **H** = 2 **M** = 1- 3/4

Injection Molded Dimension References:

G = (LAYING LENGTH) intersection of center lines to bottom of socket/thread; 90° elbows, tees, crosses; ± 1/32 inch.

H = Intersection of center lines to face of fitting; 90° elbows tees, crosses; ± 1/32 inch.

J = Intersection of center lines to bottom of socket/thread; 45° elbows; ± 1/32 inch

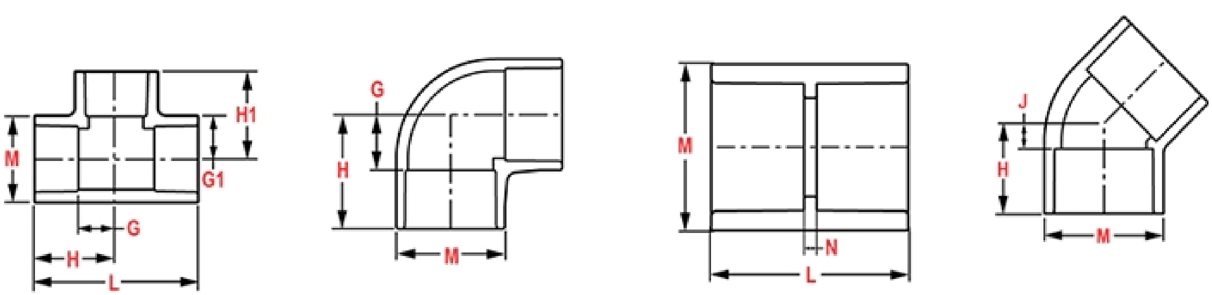
L = Overall length of fittings; ± 1/16 inch.

M = Outside diameter of socket/thread hub; ± 1/16 inch.

N = Socket bottom to socket bottom; couplings; ± 1/16 inch.

W = Height of cap; ± 1/16 inch

Typical Molded Dimension References



Spears® Technical Information

28-May-25 02:41:15 PM

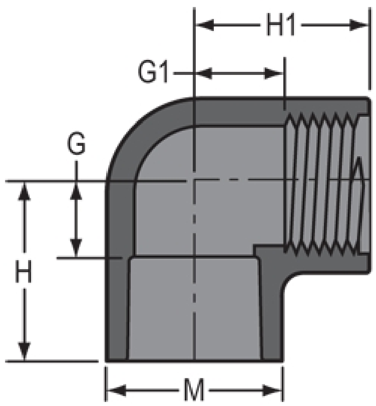
Part No: 807-010 **Schedule 80 PVC & CPVC** **Elbow**

Desc: 1 PVC 90 ELL SOCXFPT SCH80

MSRP: 36.4 **Part Code:** 080

Weight(lbs): 0.19 **Weight(kg):** 0.086 **Weight(gm):** 86

Size: 1" **Color:** GRAY **Material:** PVC



Connection Socket x Fipt

Angle 90°

Type Standard

H = 1-7/8 **H1** = 1-11/16 **M** = 1-23/32 **G** = 11/16 **G1** = 3/4

Injection Molded Dimension References:

G = (LAYING LENGTH) intersection of center lines to bottom of socket/thread; 90° elbows, tees, crosses; ± 1/32 inch.

H = Intersection of center lines to face of fitting; 90° elbows tees, crosses; ± 1/32 inch.

J = Intersection of center lines to bottom of socket/thread; 45° elbows; ± 1/32 inch

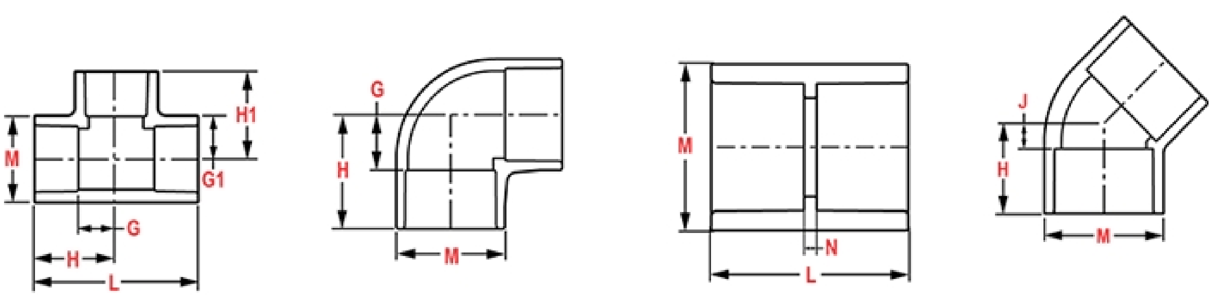
L = Overall length of fittings; ± 1/16 inch.

M = Outside diameter of socket/thread hub; ± 1/16 inch.

N = Socket bottom to socket bottom; couplings; ± 1/16 inch.

W = Height of cap; ± 1/16 inch

Typical Molded Dimension References



Spears® Technical Information

10-Sep-24 01:56:41 PM

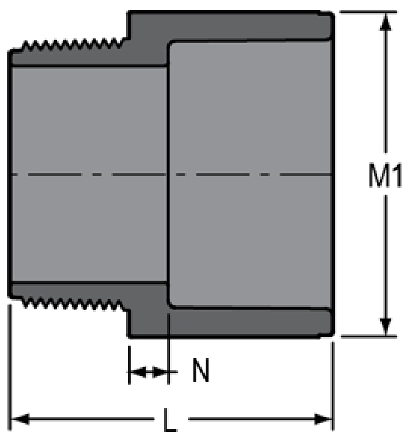
Part No: 836-010 **Schedule 80 PVC & CPVC** **Adapter**

Desc: 1 PVC MALE ADAPTER MPTXSOC SCH80

MSRP: 29.36 **Part Code:** 080

Weight(lbs): 0.093 **Weight(kg):** 0.042 **Weight(gm):** 42

Size: 1" **Color:** GRAY **Material:** PVC



Connection Mipt x Socket

Type Standard

L = 2- 5/32 **M1** = 1-23/32 **N** = 7/32

Injection Molded Dimension References:

G = (LAYING LENGTH) intersection of center lines to bottom of socket/thread; 90° elbows, tees, crosses; ± 1/32 inch.

H = Intersection of center lines to face of fitting; 90° elbows tees, crosses; ± 1/32 inch.

J = Intersection of center lines to bottom of socket/thread; 45° elbows; ± 1/32 inch

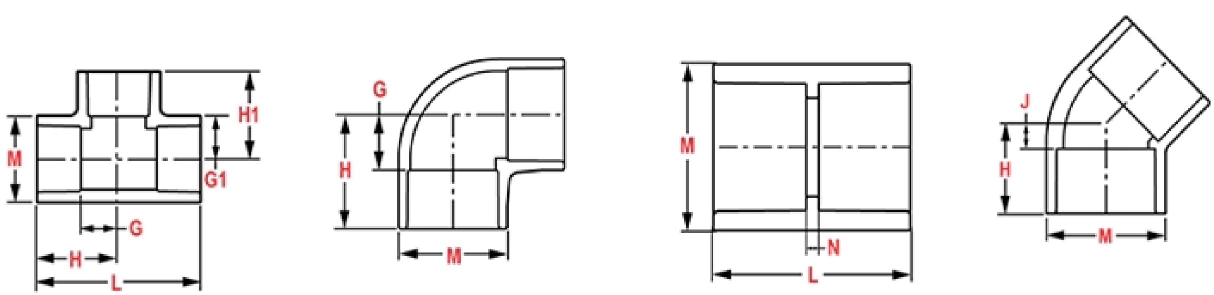
L = Overall length of fittings; ± 1/16 inch.

M = Outside diameter of socket/thread hub; ± 1/16 inch.

N = Socket bottom to socket bottom; couplings; ± 1/16 inch.

W = Height of cap; ± 1/16 inch

Typical Molded Dimension References



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TELEPHONE (818) 364-1611 • FAX (818) 364-6945 www.spearsmfg.com

CERTIFICATE OF COMPLIANCE

GRAY PVC SCHEDULE 80 MOLDED FITTINGS

Spears® Manufacturing Company certifies that its injection molded Polyvinyl Chloride (PVC) Schedule 80 fittings are manufactured in the U.S.A. in strict compliance to ASTM Standard D2467 for threaded and socket-type fittings from a PVC compound having a cell classification of 12454 conforming to ASTM Standard D1784. Fittings are certified for potable water service by NSF International®.

Alan Lunt
Director, Technical Services
alunt@spearsmfg.net

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CERTIFICATE OF COMPLIANCE

ARRA BUY AMERICAN PROVISIONS

Spears® Manufacturing Company certifies that it meets the provisions of the American Recovery and Reinvestment Act (ARRA) as defined in 48 Code Federal Regulations (CFR) Section 25.602.

For over fifty years Spears® Manufacturing Company has produced quality American made valves, fittings and pipe products sold worldwide. Spears® products are manufactured in a controlled environment within a Spears® Manufacturing facility. Spears® products are manufactured in the United States by:

Spears® Manufacturing Company
15853 Olden St.
Sylmar, CA 91342

Our commitment to the production of products solely in America qualifies our compliance to the American Recovery and Reinvestment Act (ARRA) as enacted into law on February 17th, 2009.

Doug Swingley
Chief Engineer/General Manager
dswingley@spearsmfg.net

January 2025

PVC PIPE ASSOCIATION
TECHNICAL BRIEF

PVC PIPE MATERIALS: CELL-CLASS EXPLAINED

The material used to produce PVC pipe is known as “PVC compound.” The primary ingredient, PVC resin, is blended with smaller amounts of other materials to produce the compound. These minor ingredients are used to enhance the pipe’s properties or to facilitate the manufacturing process.

PVC COMPOUND MUST MEET CELL-CLASS REQUIREMENTS

Almost all product standards for PVC water and sewer pipes require the PVC compound to meet cell-class requirements defined in ASTM D1784. The cell class consists of five cells that designate different aspects of the material. The 1st, 2nd, and 5th cells are the same for almost all PVC compounds used for municipal pipes and are summarized below:

- First cell – material = 1 for PVC pipe
- Second cell – test: IZOD impact = 2 for most PVC municipal pipe
- Fifth cell – test: deflection temperature under load = 4 for most PVC municipal pipe

THIRD & FOURTH CELLS ADDRESS TENSILE STRENGTH & TENSILE MODULUS

Tensile strength of the compound is found in the 3rd cell. Tensile strength is the design-limiting property for pressure pipe to resist hoop stress caused by internal pressure. This is less important for gravity pipe, because hoop stress is not design-limiting.

Tensile modulus of elasticity (4th cell) is the design-limiting property for non-pressure pipe to resist deflection (ovalization) caused by external loads. This is less important for pressure pipe, where external loads rarely govern wall-thickness design.

Below is part of an ASTM D1784 table that gives the values of the 3rd and 4th cells for compounds used for most municipal PVC pipes:

Cell Number	Property (Units)	Cell Limits			
		3	4	5	6
3	Tensile Strength, min. (psi)	**6,000**	**7,000**		
4	Tensile Modulus of Elasticity, min. (psi)			**400,000**	**440,000**

PRESSURE PIPE: CELL CLASS 12454

While some pressure-pipe product standards permit other cell classes, almost all PVC municipal pressure pipe is made from PVC compound meeting cell class 12454. The compound's tensile strength required is 7,000 psi minimum and tensile modulus is 400,000 psi minimum. ***(See blue highlight in table above.)***

SOLID-WALL GRAVITY PIPE: CELL CLASS 12364 OR 12454

While some gravity-pipe standards also include other cell classes, almost all PVC solid-wall gravity sewer pipe is produced from compound meeting either 12364 or 12454 cell class. For the 12364 cell class, tensile strength requirement is 6,000 psi minimum and tensile modulus is 440,000 psi minimum. ***(See green highlight in table above.)*** The 12364 cell class first appeared in ASTM PVC sewer pipe standards about 35 years ago.

“LETTER” AT THE END OF THE CELL-CLASS

Occasionally, a project specification is submitted that still contains a letter at the end of the cell-class. However, the letter should be removed from specifications, since ASTM eliminated the letter designation in a revision of the standard published in 1997. For example, the "12364-B" cell class is obsolete and should be changed to "12364."

References: ASTM D1784; *Handbook of PVC Pipe*, Uni-Bell

PVCPIPE association
Uni-Bell PVC Pipe Association | 201 E. John Carpenter Freeway, Suite 750 | Irving, TX 75062
Phone: (972) 243-3902 | Fax: (972) 243-3907 | info@uni-bell.org

Comment 5 - Ensuing Sheets

Plastic piping systems must be engineered, installed, operated and maintained in accordance with accepted standards and procedures. It is absolutely necessary that all design, installation, operation and maintenance personnel be trained in proper handling, installation requirements and precautions for installation and use of plastic piping systems before starting.

Handling & Storage

Spears® products are packaged and shipped with care to avoid damage. Pipe and fittings should be stored and protected from direct exposure to sunlight. All pipe and accessories should be stored above ground and fully supported so as not to bend or excessively deflect under its own weight. Proper stacking techniques are necessary. Improper stacking can result in instability that may result in pipe damage or personnel injury.

Use care when transporting and storing the product to prevent damage. Piping products should not be dropped or have objects dropped on them. Do not drag pipe over articles or across the ground and do not subject pipe to external loads or over stacking. If extended storage in direct sunlight is expected, pipe should be covered with an opaque material while permitting adequate air circulation above and around the pipe as required to prevent excessive heat accumulation.

Spears® products should not be stored or installed close to heat-producing sources. PVC storage should not exceed 150°F and CPVC storage should not exceed 210°F. Handling techniques for PVC and CPVC pipe considered acceptable at warm temperatures may be unacceptable at very cold temperatures. When handling pipe in cold weather, consideration must be given to its lower impact strength. In subfreezing temperatures, extra caution in handling must be taken to prevent impact damage.

All pipe should be inspected for any scratches, splits or gouges before use. Damaged sections must be cut out and discarded.

Plastic Piping Tools

Basic Tools used with Plastic Piping

Use tools that have been specifically designed for use with thermoplastic pipe and fittings when installing. A variety of tools that are designed for cutting, beveling, and assembling plastic pipe and fittings, are readily available through local wholesale supply houses dealing in plastic pipe and fittings.

•Warning Tools normally used with metal piping systems, such as hacksaws, water pump pliers, pipe wrenches, etc., can cause damage to plastic pipe and fittings. Visible and hidden fractures, scoring or gouging of material, and over tightening of plastic threaded connections are some of the common problems resulting from the use of incorrect tools and procedures.

Pipe Cutters

Pipe must be square-cut to allow for the proper joining of pipe end and the fitting socket bottom. Wheel type pipe cutters designed for plastic pipe provides easy and clean cuts on smaller pipe sizes. Care should be used with similar ratchet-type cutters to avoid damage to pipe. A slightly raised edge left on the outside of the pipe end after cutting with either device must be removed.

Pipe Cutters for Large Diameter Pipe

Blade cutters made for use with large diameter plastic pipe are easy to adjust and operate for square, burr-less cuts. Blades with carbide edges will provide longer life. With one style blade cutter, pipe ends may also be beveled for solvent joints while being cut by using an optional bevel tool in place of one cutter blade.

Hand Saws

A miter box or similar guide can be used with a fine-toothed saw blade (16 to 18 teeth per inch) having little or no set (maximum 0.025 inch).

Power Saws

Power saws are quite useful in operations where a large quantity of pipe is being cut. Blades designed for plastic pipe MUST be used. A cutting speed of 6,000 RPM, using ordinary hand pressure is recommended.

Pipe Beveling Tools

Power beveling tools, as well as hand beveling tools designed for use with plastic pipe are available. Pipe ends must be beveled (chamfered) to allow easy insertion of the pipe into the fitting and to help spread solvent cement and to prevent scraping cement from the inside of the fitting socket. A recommended bevel of 1/16" to 3/32" at a 10° to 15° angle can be achieved using a plastic pipe beveling tool, but can also be accomplished using a file designed for use on plastic.

Deburring Tools

Special plastic pipe deburring tools remove burrs from pipe ends quickly and efficiently. All burrs must be removed from the inside, as well as the outside, of the pipe ends to properly spread solvent cement when joining pipe and fitting.

Strap Wrenches

Strap wrenches with nylon straps treated for slip resistance and designed for use with plastic pipe provide gripping power for turning without scratching or deforming the pipe.

Chain Vises

Chain vises can be used to hold pipe. Vises made with jaws engineered for use with plastic pipe provide holding power without damage to the pipe.

Pullers & Joining Devices

Pipe and fitting pullers are available for joining large diameter plastic pipe and fittings. These tools are designed to allow the pipe to be inserted to the proper insertion depth, maintain proper alignment during assembly, and hold freshly solvent-cemented connections to prevent the fitting from backing-off until the initial set time is achieved.

Joining Methods -Solvent Cement Welding

Solvent cement welding is the most widely used joining method for PVC and CPVC pipe and fittings. Other methods such as threads, flanges and groove adapters can also be used. These are specifically useful where it is anticipated that the joint will have to be disassembled in the future.

Solvent Cement Safety Precautions

Solvent cement products are flammable and contain chemical solvents. Appropriate safety precautions must be taken BEFORE APPLYING PRIMER AND CEMENT. Read the cement can label!

•CAUTION

Virtually all solvent cements and primers for plastic pipe are flammable and should not be used or stored near heat, spark or open flames. Do not smoke during use. Eliminate all ignition sources. Primer and PVC cement should be stored in closed containers in the shade at temperatures between 40°F and 110°F; CPVC cement at temperatures between 40°F and 90°F. Use of a can with applicator attached to its lid is recommended. ***Verify expiration dates stamped on cements and primers prior to use.***

Avoid breathing vapors. *They should be used only with adequate ventilation. Explosion-proof general mechanical ventilation is recommended. In confined or partially enclosed areas, a ventilating device should be used. Containers should be kept tightly closed when not in use, and covered as much as possible when in use.*

Avoid contact with skin and eyes. May be absorbed through the skin; wearing PVA coated protective gloves and an impervious apron are recommended. May cause eye injury. Use eye protection and avoid eye contact. In case of contact, flush with plenty of water for 15 minutes. If irritation persists, get medical attention. If swallowed, call a physician immediately and follow precautionary statement given on side panel of cement container. Keep out of reach of children.

Refer to Solvent Cement Safety Data Sheet (SDS)

Use Caution with Welding Torches or other equipment where sparks might be involved at construction sites where plastic pipe has recently been solvent welded. Flammable vapors from cemented joints can stay within a piping system for some time. In all cases, lines should be flushed and purged to remove solvent vapors before welding.

Use Caution with Calcium Hypochlorite. Do not use a dry granular calcium hypochlorite as a disinfecting material for water purification in potable water piping systems. Granules or pellets of calcium hypochlorite (including their vapors) may react violently with solvent cements and primers if a water solution is not used. Chlorinated water solutions are nonvolatile and may be pumped into the piping system. Dry granular calcium hypochlorite should not be stored or used near solvent cements or primers.

Actually, solvent cementing is no more dangerous than putting gasoline in your automobile.

Solvent Cement and Primer Spills

Protect work areas prior to starting by using drop cloths in the event of a spill. Accidental spills should be wiped up immediately before the cement sets. Cement and/or primer spills can cause irreparable damage depending on the type of surface affected. Consult the manufacturer of the affected surface for possible suggestions.

Basic Solvent Cement Joints

The following is a general description of basic techniques used to make solvent cement joints. Adjustments will need to be made to method and tools used according to size of piping, but the same principles apply. Additional guidance can be found in ASTM D 2855, Standard Practice for Making Solvent-Cemented Joints with Poly (Vinyl Chloride) (PVC) Pipe and Fittings. **Important:** Installers should verify that they can make satisfactory joints under varying conditions and should receive training in installation and safety procedures.

To consistently make good joints in PVC and CPVC piping products, the following should be carefully understood:

1. The joining surfaces of pipe and fitting must be softened and made semi-fluid.

2. Sufficient cement must be applied to fill the gap between pipe and fitting.

3. Assembly of pipe and fittings must be made while the surfaces are still wet and fluid.

4. Joint strength develops as the cement dries (cures). In the tight part of the joint (interference area) the surfaces will fuse together; in the loose part the cement will bond to both surfaces.

Cutting the Pipe

PVC or CPVC pipe can be cut easily with a ratchet cutter, wheel-type plastic pipe cutter (**NOTE:** be sure to remove any raised ridge produced by wheel cutters), a power saw, or any other fine-tooth saw. It is important that the cutting tools being used are designed for plastic pipe. To ensure that the pipe is cut square, use a miter box when cutting with a saw. Cutting pipe as square as possible provides the maximum bonding surface area.

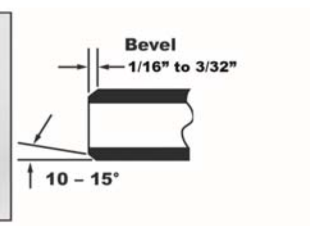
Be careful not to split the tube if using a ratchet-type cutter, especially in temperatures below 50°F. If any damage or cracking is evident, cut off at least 2" of the pipe beyond any visible crack.

Deburring & Beveling

Burrs and filings can prevent contact between the tube and the fitting during assembly and must be removed from the outside and the inside of the pipe. A deburring/chamfering tool (or file) is suitable for this purpose:

Burrs Being Removed from Outside & Inside

A slight bevel (chamfer) must be placed at the outside end of the pipe to ease the entry of the tube into the socket and minimize the chance of cement being wiped off the fitting:



Bevel Outside End

Fitting & Joining Preparation

1. Using a clean, dry rag, wipe any loose dirt and moisture from the fitting's socket and pipe end. Moisture can slow the cure time, and at this stage of assembly, excessive moisture can reduce joint strength.

2. Check the dry fit of the pipe and fitting. The pipe should enter the fitting's socket easily 1/3 - 2/3 of the way (interference fit) , or at least have interference between pipe and fitting bottom (net fit). **DO NOT** use any components that appear irregular or do not fit properly. Contact Spears® regarding any questions about usability.

3. Measure socket depth and mark on pipe for reference during cement application.

4. It is advisable to additionally mark pipe and fitting for alignment orientation position, especially with larger fittings.

Solvent Cementing Assembly

Verify the expiration date located on the solvent cement can. The cement can be used for a period of 2 years from the date stamped on the can. When cementing pipe and fittings in extremely cold temperatures, make sure the cement has not "JELLED." Jelled or expired cement must be discarded in an environmentally friendly fashion, in accordance with local regulations. To prolong the life of solvent cement, keep the containers tightly closed when not in use, and cover the container as much as possible during use. If an unopened solvent cement container is subjected to freezing temperatures, the cement may become extremely thick. Place the closed container in a room temperature area where, after a short time period, the cement will return to a usable condition. **DO NOT** attempt to heat solvent cement. The cement must be applied when the pipe and fittings are clean and free from any moisture and debris.

Primer Use - Softening of pipe and fitting joining surfaces can be achieved by the cement itself or by using a suitable primer. A primer will usually penetrate and soften the surfaces more quickly than the cement alone. However, special "one-step" cements formulated for use without primers are available. Check local codes (where required) for acceptable applications.

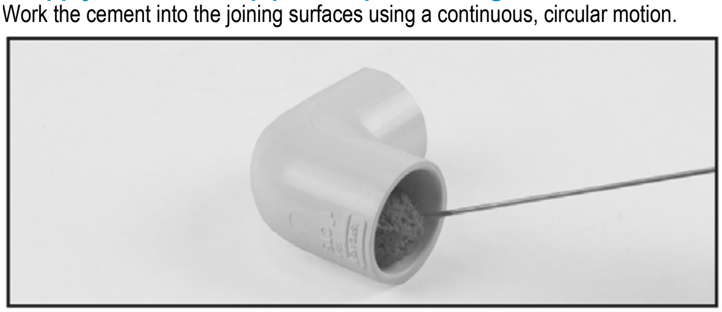
Apply Primer - USING AN APPLICATOR THAT IS AT LEAST 1/2 THE SIZE OF THE PIPE DIAMETER, vigorously scrub joining surface of fitting, of pipe and then again of fitting. Work quickly to apply 2-3 coats in this manner. SOLVENT CEMENT SHOULD THEN BE APPLIED WHILE PRIMER IS STILL WET.

Apply Solvent Cement - USING AN APPLICATOR THAT IS AT LEAST 1/2 THE SIZE OF THE PIPE DIAMETER, WORK THE CEMENT INTO THE JOINING SURFACES USING A CONTINUOUS, CIRCULAR MOTION.

Use sufficient cement, but avoid puddling the cement on or within the fitting and pipe. Puddled cement causes excess softening and damage to the PVC or CPVC material. If interference fit was at the bottom of the socket, use extra cement and make a 2nd application to pipe. WORK QUICKLY SO THAT PIPE AND FITTING CAN BE JOINED WHILE CEMENT IS STILL WET.

Apply the cement in the sequence pictured below:

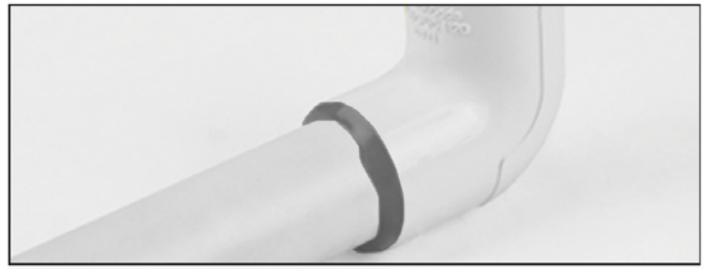
1. Apply a coat to the pipe to depth of fitting socket
Work the cement into the joining surfaces using a continuous, circular motion.



2. Apply a medium coat to the fitting socket
Avoid puddling the cement in the sockets and avoid getting cement in other sockets or threaded connections.
3. Apply a second coat to the pipe end for sizes 1-1/4 inch and larger joints, or if interference fit was at socket bottom during dry-fit.

Assemble Joint

Immediately insert pipe into the fitting socket while rotating the pipe 1/4 turn. Align the fitting in the proper orientation at this time. Make sure the pipe bottoms out at the fitting's stop. Hold the assembly for at least 30 seconds to ensure initial bonding. Tapered pipe sockets can result in pipe backing out of the joint if not held under constant pressure A bead of cement must be present around the pipe and fitting juncture. If this bead is not continuous around the socket's shoulder, insufficient cement was applied and the joint must be disassembled or cut out and replaced.



Any cement, in excess of the bead, can be wiped off with a dry, clean rag.

Set and Cure Times

SET TIME: The initial set time is the recommended waiting period before handling newly assembled joints. After initial set, the joints will withstand the stresses of normal installation. However, a badly misaligned installation will cause excessive stresses in the joint, pipe and fittings.

CURE TIME: The cure time is the recommended waiting period before pressurizing newly assembled joints.

The following basic guidelines should be used:

• The set and cure times for solvent cement depend on pipe size, temperature, relative humidity, and tightness of fit. Drying time is faster for drier environments, smaller pipe sizes, high temperatures, and tighter fits.

• Special care must be taken when assembling products in low temperatures (below 40°F) or high temperatures (above 80°F).

• Extra set and handling times must be allowed in colder temperatures. When cementing pipe and fittings in cold temperatures, make sure the cement has not "JELLED." Jelled cement must be discarded.

• In higher temperatures, make sure both surfaces to be joined are still wet with cement during assembly.

• The assembly must be allowed an initial set, without any stress on the joint

• Following the initial set period, the assembly can be handled carefully by avoiding stress on the joint.

Average Set Times

Temp. Range	Pipe Sizes 1/2"- 1-1/4"	Pipe Sizes 1-1/2"- 2"	Pipe Sizes 2-1/2"- 8"	Pipe Sizes 10"- 15"	Pipe Sizes 16"- 24"
60° - 100°F	2 Min.	5 Min.	30 Min.	2 Hrs.	4 Hrs.
40° - 60°F	5 Min.	10 Min.	2 Hrs.	8 Hrs.	16 Hrs.
0° - 40°F	10 Min.	15 Min.	12 Hrs.	24 Hrs.	48 Hrs.

Suitable for Oil-Free air handling to 25 psi, not for distribution of compressed air or gas
See Spears® Product Sourcebook for product offerings

Average Cure Times

Relative Humidity 60% or Less*	Pipe Sizes 1/2" - 1-1/4"		Pipe Sizes 1-1/2" - 2"		Pipe Sizes 2-1/2" - 8"		Pipe Sizes 10" - 15"	Pipe Sizes 16" - 24"
Temperature Range During Assembly and Cure Periods	Up to 160 psi	Above 160 to 370 psi	Up to 160 psi	Above 160 to 315 psi	Up to 160 psi	Above 160 to 315 psi	Up to 100 psi	Up to 100 psi
60° - 100°F	15 Min.	6 Hrs.	30 Min.	12 Hrs.	1-1/2 Hrs.	24 Hrs.	48 Hrs.	72 Hrs
40° - 60°F	20 Min.	12 Hrs.	45 Min.	24 Hrs.	4 Hrs.	48 Hrs.	96 Hrs.	6 Days
0° - 40°F	30 Min.	48 Hrs.	1 Hr.	96 Hrs.	72 Hrs.	8 Days	8 days	14 Days

•NOTE In damp or humid weather allow 50% more cure time. The cure schedules shown are suggested as guides only. They are based on laboratory test data, and should not be taken to be the recommendations of all cement manufacturers. Individual solvent cement manufacturer's recommendations for the particular cement being used should be followed.

Special Considerations for Working with Solvent Cement Welding

Handling of Cement

Keep cement containers covered while not in use. Cement with the lid left off can become thick and viscous, or gel like. This condition is typically a result of tetrahydrofuran (THF) solvent evaporation and the cement is useless. Do not try to restore the cement by stirring in a thinner. Smaller containers of cement are recommended to be used, especially in warm or hot weather. Prior to opening cans of cement, shake vigorously to properly mix resin and solvents. Solvents contained in PVC and CPVC cements are highly flammable and should not be used near an open flame. The area in which the cement is being used should be well ventilated, and prolonged breathing of the fumes should be avoided, as well as contact with the skin or eyes. Verify the expiration dates stamped on the cements and primers prior to use.

CEMENT AND PRIMER SHELF LIFE

Spears® Products	Shelf Life	Spears® Products	Shelf Life
Primers / Cleaners	3 years	CPVC Solvent Cement	2 years
PVC Solvent Cement	3 years	ABS Solvent Cement	3 years

Hot Weather Use

Problems can be avoided when solvent cementing in 95°F or higher temperatures by taking a few special precautions. Solvent cements evaporate faster at elevated temperatures and can dry out prematurely. This is especially true when there is a hot wind blowing. Dry cement on pipe or fitting socket prior to assembly will not bond. If the pipe has been in direct sunlight for any length of time, surface temperatures may be 20°F to 30°F above air temperature. Solvents attack these hot surfaces faster, deeper and dry out quicker. As a result, it is very important to avoid puddling inside sockets, assemble immediately while wet and to wipe off excess cement at the joint exterior.

Tips for Solvent Cementing in High Temperatures:

1. Store solvent cements in a cool or shaded area prior to use.

2. If possible, store the fittings and pipe, or at least the ends to be solvent welded, in a shady area before cementing.

3. Cool surfaces to be joined by wiping with a damp rag. HOWEVER, be sure that surfaces are dry prior to applying solvent cement.

4. Try to do the solvent cementing in cooler morning hours.

5. Make sure that both surfaces to be joined are still wet with cement when putting them together.

Cold Weather Use

Solvent Cements and primers have excellent cold weather stability and are formulated to have well balanced drying characteristics even in subfreezing temperatures. Good solvent cemented joints can be made in very cold conditions provided proper care and a little common sense are used. In cold weather, solvents penetrate and soften surfaces more slowly than in warm weather. The plastic is also more resistant to solvent penetration, therefore, it becomes more important to pre-soften surfaces. A longer cure time is necessary due to slower evaporation.

Tips for Solvent Cementing in Cold Temperatures:

1. Prefabricate as much of the system as possible in a heated work area.

2. Store cements in a warmer area when not in use and make sure they remain fluid.

3. Take special care to remove moisture, including ice and snow.

4. Use special care to ensure joining surfaces are adequately softened; more than one application may be necessary.

5. Allow a longer cure period before the system is used.

Effects of Tolerances and Fits

PVC pipe and fittings are manufactured to applicable ASTM Standards to produce an interference fit when assembled. However, minimum and maximum allowable tolerances permitted for pipe and fitting can result in variations. For example, fitting with the maximum diameter and the pipe with the minimum diameter, may result in a loose fit. Applying two coats of solvent cement will help assure a good joint. Conversely, if the pipe diameter is on the maximum side and the fitting on the minimum side, the interference may be too great and sanding of the pipe O.D. may be necessary to permit entrance.

Always check dry fits prior to making a joint. If fit is loose, multiple coats and use of an extra heavy bodied cement may be required. Mating components should be checked to assure that tolerances and engagements are compatible (see preceding Basic Solvent Cement Joints instructions). Inspect all pipe and fittings for damage or irregularities. Do not use any components that appear irregular or do not fit properly. Contact the appropriate manufacturer of the product in question to determine usability.

Large Diameter Pipe

Basic Solvent Cement Joint instructions apply to all sizes of pipe, but when making joints larger than 4", the use of two persons is recommended to properly apply cement and immediately assemble the joint while the cemented surfaces are still wet. Alignment of large diameter pipe and fittings during joining is critical since there is a greater potential for movement in the upper portion of a tapered socket that can result in misalignment. Special tools are commercially available for joining large diameter pipe.

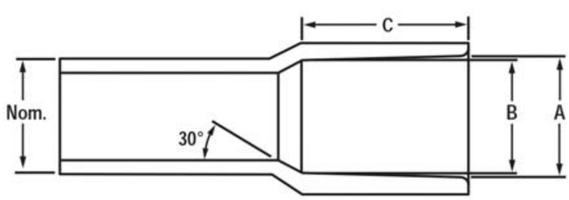
Be sure to use an appropriate size roller applicator with large diameter pipe, along with a heavy or extra-heavy bodied cement that is medium to slow setting. These have increased gap filling capability and allow somewhat longer assembly time. However, applications of heavy coats of solvent cement and speed in making the joint is important. Under a damp or wet condition, solvent cement may absorb some moisture. Excessive moisture can slow down the cure and reduce joint strength. Spears® CPVC-24 heavy body or PVC-19 extra-heavy body solvent cements are excellent for joining large diameter pipe (see Solvent Cement Selection Guide in following sections).

Belled End Pipe

Commercially available belled end pipe can be used to eliminate the need for couplings. Where belled end pipe is used, it is suggested that the interior surface of the bell be penetrated exceptionally well with the primer.

•NOTE some manufacturers use a silicone release agent on the belling plug, and a residue of this agent can remain inside the bell. Silicone will contaminate the joint and not allow proper solvent cement welding. All silicone residue must be removed in the cleaning process prior to solvent cementing.

Belled-End Pipe Dimensions



Nominal Size (in.)	A		B		C.
	Min.	Max.	Min.	Max.	Min.
1-1/4	1.675	1.680	1.648	1.658	1.870
1-1/2	1.905	1.914	1.880	1.888	2.000
2	2.381	2.393	2.363	2.375	2.250
2-1/2	2.882	2.896	2.861	2.875	2.500
3	3.508	3.524	3.484	3.500	3.250
4	4.509	4.527	4.482	4.500	4.000
5	5.573	5.593	5.543	5.563	4.000
6	6.636	6.658	6.603	6.625	6.000
8	8.640	8.670	8.595	8.625	6.000
10	10.761	10.791	10.722	10.752	8.000
12	12.763	12.793	12.721	12.751	8.500
14	14.030	14.045	13.985	14.000	9.000
16	16.037	16.052	15.985	16.000	10.000
18	18.041	18.056	17.985	18.000	12.000
20	20.045	20.060	19.985	20.000	12.000
24	24.060	24.075	24.000	24.015	14.000

Estimated Quantities of Solvent Cement

A variety of conditions can affect the amount of solvent cement required for making reliable joints. These include pipe size, tolerances, socket depths as well as installation conditions and type of cement used. Fitting sockets are tapered for proper assembly, which produces a slight gap at the socket entrance when installed with pipe. As pipe sizes increase, heavier bodied cements should be used for increase gap filling capabilities. It is best to use liberal amounts of solvent cement since insufficient cement use is one of the most common reasons for joint failure. The following information on cement usage is a recommendation only and other factors or unanticipated conditions may be encountered. Quantities are based on use with average socket lengths of Spears® molded and fabricated fittings.

Standard Pipe Joints

Fitting Size (in.)	Joints per Pint*	Joints per Quart*	Joints per Gallon*
1/2	150	300	1200
3/4	100	200	800
1	63	125	500
1-1/4	54	108	432
1-1/2	45	90	360
2	30	60	240
2-1/2	25	50	200
3	20	40	160
4	15	30	120
6	5	10	40
8	3	5	20
10	---	2-3	4-6
12	---	1-2	2-4

* The number of joints listed are an estimation and should be used as a general guideline only.

Large Diameter Pipe Joints

Fitting Size (in.)	Joints per Quart*	Joints per Gallon*
14	1	4
16	5/8	2-1/2
18	1/2	2
20	7/16	1-3/4
24	5/16	1-1/4

* The number of joints listed are an estimation and should be used as a general guideline only.

Supplemental Information on Solvent Cementing

Applicators

A wide variety of daubers, brushes, and rollers are available. For proper solvent cement welding of pipe and fittings, the cement applicator must be no less than half the size of the pipe. Sufficient cement cannot be applied using daubers attached to the cement can lid on large diameter products (> 3"dia.) The following chart shows a variety of Spears® applicator sizes for use on different pipe diameters.

SPEARS® APPLICATOR SELECTION GUIDE

For proper solvent cement welding of pipe and fittings, the cement applicator must be no less than half the size of the pipe

DAUBERS	Pipe Diameters						
	1/4"	1/2"	3/4"	1"	1-1/4"	1-1/2"	2"
3/8" Dauber	•	•	•				
1/2" Dauber			•	•			
3/4" Dauber					•	•	
1-1/4" Dauber							•

ROLLERS & SWABS	SIZE	FOR PIPE DIAMETERS
3020	3" Roller	3" - 6"
6020	4" Roller	3" - 8"
7020	7" Roller	6" +
5520	4" Roller	6" +
6520	6" Roller	6" +
4020	4" Swab	6" +
5020	4" Swab	3" - 8"
4520	4" Swab	6" +

Cleaners

Cleaners can be used to remove dirt, oil and grease from the bonding surfaces of PVC, CPVC, ABS and Styrene pipe and fittings. Use of a cleaner is recommended before priming of pipe and fittings.

Primers

The use of Primer is necessary to penetrate and dissolve the surface of the pipe and fitting prior to the application of cement. Special "one-step" cements formulated for use without primers are available. Check cement instructions and local codes (where required) for acceptable applications. Primer must be applied to both the pipe and fittings. Apply multiple coats of primer to the fitting socket and to the outside of the pipe ensuring that the entire surface is wet. Solvent cement must be applied immediately after primer while the surfaces are still tacky.

Solvent Cements

Solvent cements are produced for joining a variety of commercially available pipe and fitting materials, including PVC, CPVC and ABS plastics. Solvent cements are typically formulated using tetrahydrofuran (THF). When properly applied, this solvent dissolves the mating surfaces of the pipe and fittings. Cyclohexanone is a typical retardant used to slow the rate of solvent evaporation. Immediate joining of the wet mating surfaces in one minute or less is essential to eliminate dry spots that will not bond. The bond interface is strongest at the area of interference fit where the softened semi-fluid surfaces of the pipe and fitting chemically fuse. Plastic resin fillers (dissolved PVC or CPVC) in the cement fill the gaps between pipe and fitting. Cements are available in clear, white, gray and other colors to match the pipe or for specific application. **Inert pigments are used for coloration.** For example, white cements are made from titanium dioxide while gray cements are made from titanium dioxide and carbon black. As the solvent evaporates, pipe and fitting joint "cures", except for some residual solvent that dissipates over time. The resulting fused area is why this method is called "solvent cement welding" although no heat is applied to melt and fuse the bonded areas as in metal welding.

Solvent cements are formulated in regular bodied, medium bodied, heavy bodied, extra heavy bodied and specialty cements. Different types of cements have different set and cure times. Low VOC products - with lesser VOC emissions - will contribute to cleaner air and better workplace conditions. All Spears® solvent cement and primer products are certified as Low VOC.

1. **Regular Bodied -** Cements for smaller diameters (i.e.< 4") and thin-wall classes and Schedule 40 piping with interference fits. Generally referred to as "regular body" such as Spears® PVC-00 and PVC-02 cements, these cements are fast setting.

2. **Medium Bodied -** Cements for smaller diameters (i.e. < 4") for all classes, Schedule 40 and Schedule 80 pipe with interference fits such as Spears® PVC-05 and PVC-21 cements. These cements have better gap filling capability than regular bodied cement and are also considered fast setting

3. **Heavy Bodied & Extra Heavy Bodied -** Cements for both small and large diameters of heavier-wall Schedule 80 and Schedule 120 products. Heavy-body such as Spears® PVC-11 and CPVC-24 cements are classified as medium setting and extra heavy-body such as Spears® PVC-19 cement is classified as slow setting. These cements are formulated to fill larger gaps, dry slower and typically take longer to dry in order to provide more time to assemble joints.

4. **Specialty Cements -** Specialty cements formulated for use with specific products and applications, but can also be used with other applications of similar products . Examples include special cements such as Spears® PVC-25 Wet-N-Dry; transition cements such as Spears® MULTIPURPOSE-90 and Spears® ABS TO PVC-94; product specific cements such as Spears® ABS-71 and ABS-73; and one-step specialty cements. One-step cements do not require the use of primer prior to the application of the cement. Examples include Spears® FS-5 one-step cement for use with FlameGuard® CPVC Fire Sprinkler Products, Spears® LW-4 one-step cement for use with **LabWaste®** CPVC Chemical Drainage Systems; Spears® **EverTUFF®** CTS-5 for use with CPVC hot and cold water plumbing systems, and Spears® LX-5 Low Extractable PVC cement for use in high purity applications (i.e. Spears® LOW EXTRACTABLE PVC products). In addition, special application cements such as Spears® CPVC-24 is formulated for improved chemical resistance to caustics and chemical applications with both PVC and CPVC products. In fact, CPVC-24 is one of the most versatile solvent cements on the market today.

Selecting the appropriate solvent cement and primer for the type of products being joined is important. The following selection guide can be used in selecting the right Spears® solvent cement and primer for your application.

Spears® Solvent Cement & Primer Selection Guide

Type Body		Spears®/IPS Cross Reference		Color	Relative Set	Capacity	Features
		Spears®	IPS				
PVC	Regular	PVC-00	700	Clear	Fast	Schedule 40 - 4"	
PVC	Regular	PVC-02	702	Clear	Fast	Schedule 40 - 4"	Dries Clearest, slightly thicker than PVC-00
PVC	Medium	PVC-05	705 (Not in Gray)	Clear/Gray	Fast	All Classes & Schedule 40 - 6" Schedule 80 - 4"	Industrial Duty Primerless Capability[1]
PVC	Medium	PVC-21	721	Blue	Fast	Schedule 40 - 6" Schedule 80 - 4"	Primerless Capability[1]
PVC	Heavy	PVC-11	711	Gray	Medium	All Classes & Schedules - 12"	Industrial Duty
PVC	Heavy	PVC-17	717	Clear/Gray	Medium	All Classes & Schedules - 12" Non-pressure - 18"	Industrial Duty
PVC	Extra Heavy	PVC-19	719	White/ Gray	Slow	All Classes & Schedules Requiring High Gap Filling - 30"	Industrial Duty
PVC	Specialty Cements	PVC-25	725	Aqua Blue	Very Fast	All Classes & Schedule 40 - 6" Schedule 80 - 4"	Wet-N-Dry Formulation Primerless Capability[1]
PVC	Specialty Cements	PVC-26	747	Blue	Very Fast	All Classes & Schedule 40 - 6" Schedule 80 - 4"	POOL-PRO™ Formulation for Pool & Spa Primerless Capability[1] Fades to clears as it cures.
PVC	Specialty Cements	PVC-27	727	Clear	Very Fast	All Classes & Schedule 40 - 6" Schedule 80 - 4"	Cold-N-Hot Formulation -15°F to 110°F Primerless Capability[1]
PVC	Specialty Cements	PVC-37	737	Blue	Very Fast	All Classes & Schedules - 6" Schedule 80 - 4"	Formulated for Wet Conditions Primerless Capability[1]
PVC	Specialty Cements	PVC-50	750	Blue	Very Fast	Schedule 40 - 6" Schedule 80 - 4"	HOT PVC Formulation Primerless Capability[1]
PVC	Specialty Cements	PVC-95	795	Clear/ Blue	Fast	All Classes & Schedule 40 - 6" Schedule 80 - 4"	Flexible PVC for Flex-Flex, Flex-Rigid Joints
PVC	Specialty Cements	LX-5	N/A	Clear	Fast	Low Extractable PVC Systems - 6"	One Step[2] Low Extractable, High Purity Cement
CPVC	Heavy	CPVC-24	724	Gray/ Orange	Medium	All Classes & Schedules PVC or CPVC- All Sizes	both CPVC & PVC Systems - including Duct - Industrial Duty
CPVC	Specialty Cements	LW-5	N/A	Mustard	Medium	LabWaste® CPVC Drainage Systems - 24"	One Step[2] Cement-Only LabWaste® System Chemically Approved Cement
CPVC	Specialty Cements	FS-5	N/A	Red	Fast	CPVC Fire Sprinkler Systems - 3"	One-Step[2] Cement for all CPVC Fire Sprinkler Systems
CPVC	Specialty Cements	CTS-5	FlowGuard Gold	Yellow	Fast	CTS CPVC Systems" - 2	One-Step[2] Cement for all CTS CPVC Systems
ABS	Medium	ABS-71	771	Yellow/ Milky	Fast	All ABS Classes & Schedules - 8"	
ABS	Medium	ABS-73	773	Black	Fast	All ABS Classes & Schedules - 8"	
Transition & Multipurpose	Medium	ABS TO PVC-94	794	Green	Fast	All Classes & Schedules - 6" (Except Schedule 80)	For ABS-to-PVC Transition Joints
Transition & Multipurpose	Medium	MULTIPURPOSE-90	790	Clear	Fast	All Classes & Schedules - 6" Schedule 80 - 4"	For PVC and CPVC pressure ABS and Styrene low-pressure systems
Primers & Cleaners		Primer-75	P-75	Aqua Blue		All Classes, Schedules & Sizes PVC & CPVC	Formulated for Wet Conditions Industrial Duty
Primers & Cleaners		Primer-70	P-70	Purple/ Clear		All Classes, Schedules & Sizes PVC & CPVC	Industrial Duty
Primers & Cleaners		Primer-68	P-68	Purple/ Clear		All Classes, Schedules & Sizes PVC & CPVC	
Primers & Cleaners		Cleaner-65	C-65	Clear		All Classes, Schedules & Sizes PVC & CPVC	For PVC, CPVC, ABS or Styrene
Primers & Cleaners		Primer Cleaner-64	PC-64	Purple		All Classes, Schedules & Sizes PVC & CPVC	

Notes
1= Primerless Capability indicates a cement can be used without primer in certain applications if local code permits. See specific cement information for further restrictions.
2 = One Step designates a cement specicifically designed for use without primer. CTS One Step acceptability depends on local code requirements.
CTS = Copper Tube Size, ASTM D2846 Hot and Cold Water Distribution System.

Suitable for Oil-Free air handling to 25 psi, not for distribution of compressed air or gas
See Spears® Product Sourcebook for product offerings

Joining Method - Threaded Connections

Threaded connections require the application of a thread sealant that is compatible with PVC and CPVC material. Spears® recommends the use of Spears® Blue 75™ Thread Sealant.

CAUTION - Use only thread sealants recommended for PVC or CPVC plastic. Other joint compounds or pastes may contain substances that could cause stress cracks in PVC or CPVC materials.

Apply sealant to the male threads only. Make sure all threads are covered. **DO NOT** clog the waterway with excess sealant. If PTFE tape must be used, Spears® recommends a thickness of at least .0035" that meets or exceeds military specification, MIL-T-27730A. **DO NOT** use a combination of tape and thread sealant on the same joint. Apply PTFE tape in the direction of the threads by starting with the first full thread and continuing over the entire thread length. Make sure all threads are covered. Generally, 2 - 3 wraps are sufficient to produce a watertight connection

DO NOT over-torque any threaded connections. Generally, one to two turns beyond finger-tight are required for a threaded connection. Use a smooth-jawed wrench or strap wrench when installing threaded connections.

Threading Pipe

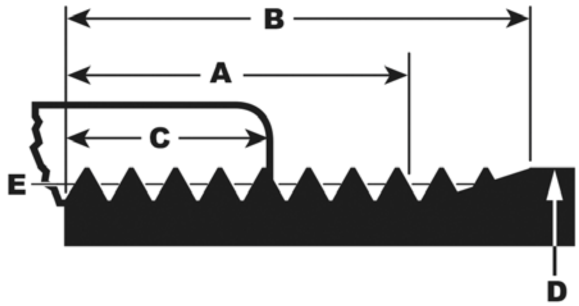
PVC and CPVC pipe can be threaded using either standard hand pipe stocks or power-operated equipment. Since rigid PVC plastic pipe has the same outside diameter as standard steel pipe in comparable sizes, standard steel pipe taps and dies can be used. A cut thread or deep scratch results in a stress concentration point. As a result, only Schedule 80 and Schedule 120 pipe should be threaded. A 50% pressure de-rating is applied to threaded pipe to compensate for this. **DO NOT** thread Schedule 40 pipe. For optimum results in threading, use new taps and dies; but in any case, they should be cleaned and sharpened and in good condition. Power threading machines should be fitted with dies having a 5° negative front rake and ground especially for this type of pipe; tapered guide sleeves are not required. For hand stocks the dies should have a negative front rake of 5° to 10°. Dies which have been designed for use on brass or copper pipes may be used successfully. Carboloy dies give longer service. (Taps should be ground with a 0° to 10° negative rake, depending upon the size and pitch of the thread. Die chasers should have a 33° chamfer on the lead; a 10° front or negative rake; and a 5° rake on the back or relief edge.). Self-opening die heads and collapsible taps, power threading machines and a slight chamfer to lead the tap or dies will speed production; however, taps and dies should not be driven at high speeds or with heavy pressure.

A tapered plug should be inserted into the pipe when threading, to hold the pipe round and to prevent the die from distorting and digging into the pipe wall. This ensures uniform thread depth all the way around. Pipe for threading should be held in a suitable pipe vise, but saw-tooth jaws should not be used. Flanges and close nipples should be threaded in jigs or tapping fixtures. To prevent crushing or scoring the pipe, some type of protective wrap, such as canvas, emery paper, or a light metal sleeve should be used; rounding of chuck jaws will also be helpful. Rigid PVC or CPVC plastic pipe should be threaded without use of lubricants. Standard cutting oils can cause stress cracking in plastics and should not be used. Water-soluble oil or plain water is recommended. Degreasing with any solvents is not recommended, nor should solvents be used in any cleanup. Always clear cuttings from the die.

DO NOT OVER THREAD - To obtain a tight, leak proof joint, the thread dimensions shown in the table should be used. If pipe is over threaded, fittings cannot be run on far enough to make a tight seal.

American National Standards Institute Code B1.20.1 covers dimensions and tolerances for tapered pipe threads. **Only Schedule 80 or heavier wall pipe should be threaded.**

Angle between sides of thread is 60 degrees. Taper of thread, on diameter, is 3/4 inch per foot. The basic thread depth is 0.8 x pitch of thread and the crest and root are truncated an amount equal to 0.033 x pitch, excepting 8 threads per inch which have a basic depth of 0.788 x pitch and are truncated 0.045 x pitch at the crest and 0.033 x pitch at the root.



PIPE THREADS

Nominal Size (in.) (Max.) (In.)	Outside Diameter (in.) D	Number of Threads Per Inch	Normal Engagement By Hand (in.) C	Length of Effective Thread (in.) A	Total Length: End of pipe to vanish point (in.) B	Pitch Diameter at end of Internal Thread (in.) E	Depth of Thread (Max.) (in.)
1/8	0.405	27	0.180	0.2639	0.3924	0.37476	0.02963
1/4	0.540	18	0.228	0.4018	0.5946	0.49163	0.04444
3/8	0.675	18	0.240	0.4078	0.6006	0.62701	0.04444
1/2	0.840	14	0.320	0.5337	0.7815	0.77843	0.05714
3/4	1.050	14	0.339	0.5457	0.7935	0.98887	0.05714
1	1.315	11-1/2	0.400	0.6828	0.9845	1.23863	0.06957
1-1/4	1.660	11-1/2	0.420	0.7068	1.0085	1.58338	0.06957
1-1/2	1.900	11-1/2	0.420	0.7235	1.0252	1.82234	0.06957
2	2.375	11-1/2	0.436	0.7565	1.0582	2.29627	0.06957
2-1/2	2.875	8	0.682	1.1375	1.5712	2.76216	0.10000
3	3.500	8	0.766	1.2000	1.6337	3.38850	0.10000
4	4.500	8	0.844	1.3000	1.7337	4.38713	0.10000
5	5.563	8	0.937	1.4063	1.8400	5.44929	0.10000
6	6.625	8	0.958	1.5125	1.9462	6.50597	0.10000

Joining Method - Flanged Connections

PVC and CPVC flanges are available in several designs, including solid one-piece flanges, two-piece Van Stone style flanges featuring a movable ring for bolt alignment, and blind flanges for capping off a piping run. CPVC and PVC flanges are available in socket, spigot and threaded configurations and are coupling devices designed for joining IPS (Iron Pipe Size) plastic piping systems where frequent disassembly may be required. Flanges can be used as a transitional fitting for joining plastic to metal piping systems, and for connection to other flanged type valves and equipment. A flat gasket is used between flanges to form a water-tight seal. Proper gasket material should be selected for fluid compatibility. Most plastic flanges carry a maximum working pressure rating of 150 psi non-shock for water at 73° F (23° C). Pressure ratings may vary according to size and flange construction. Consult flange manufacturer.

Gaskets

Select appropriate size and bolt pattern gasket. Full faced, 1/8" thick elastomer flat gaskets with a Shore "A" Durometer of approximately 70 (±5) are recommended. A full-face flat gasket should cover the entire flange face but not obstruct the waterway. Verify that the gasket material is chemically compatible for use with the application fluid conveyed through the system.

Low Torque Double Convex Design Full Face Gaskets

Low torque-style gaskets require up to 50% less torque to achieve a leak-free seal. When using this type of gasket in combination with Spears® flanges be sure to follow the torque recommendations from the low torque gasket manufacturer. Over-torquing a low torque gasket may cause damage to the flange.

DO NOT USE: Spiral wound, ring, kammprofile, corrugated metal gaskets, non-asbestos or cork gaskets.

Bolt Patterns & Fastener Selection

Most PVC and CPVC flanges are produced with ANSI B16.5 bolt patterns for Class 125/150 flanges. Optional Class 300 bolt patterns (NOT a 300psi rating), certain ANSI/Metric dual pattern flanges, and metric bolt patterns can be produced. Proper bolt size, number and length should be selected for the specific flanges and equipment being assembled. Bolt length requirements will vary according to the flange or equipment manufacturer. Spears® offers bolt packs (HK series) for use with Spears® flange to flange assemblies.

Bolts: To ensure minimal friction use a zinc-plated grade 5 bolt or stud of suitable length in accordance with ASME B18.2.1 to accommodate (2) flange thicknesses, (2) washers and (1) 1/8" full face gasket thickness.

Nuts: Use grade 5 zinc-plated hex nuts in accordance with ASME B18.2.2. Orient nut so flat side contacts flange surface.

Washers: Always use (2) wide flat washers for each bolt, one under the bolt head and one under the nut. Use flat, zinc-plated steel (Plate type) grade 5 washers in accordance with ASME B18.21.1 (Do not use thin "fender" washers).

Important: Stainless steel fasteners can be used. Be sure to use a compatible anti-seize lubricant to eliminate galling.

Bolt Preparation & Torque

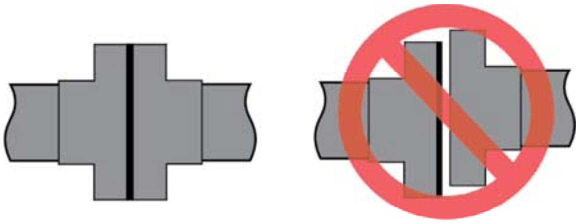
Threads should be cleaned and well lubricated. Apply a liberal amount of a chemically compatible copper-graphite anti-seize bolt lubricant to each bolt thread. Use a lubricant with a K Value (nut factor) of 0.172 ± 0.009. Testing is required to determine the exact K values and specific performance on any individual fastener using the desired chemically compatible anti-seize compound. Adequate application of thread lubricant is indicated by lubricant extrusion from the threaded joint as the nut is tightened. (**WARNING:** Use only bolt lubricants compatible with PVC or CPVC material).

Hand tighten all nuts until they are snug prior to torquing flange bolts. Using a calibrated torque wrench accurate to within ± 1ft-lb. tighten each bolt in approximately 5 ft-lb. increments using a 180° opposing sequence. Follow the numbers sequence engraved next to each Spears® flange bolt hole or consult **Torque Sequence - Table 1** information. Actual field conditions may require variations in these recommendations. **UNNECESSARY OVER-TORQUING WILL DAMAGE THE FLANGE.** Torque should always be applied in approximately 5 ft-lb. increments using a 180° opposing sequence.

Flange Make-up

Follow proper solvent cementing and/or threaded component procedures as applicable to join the flange to the pipe. Once a flange is joined to pipe, the method for joining two flanges is as follows:

1. Piping runs joined to the flanges must be installed in a straight-line position to the flange to avoid stress at the flange due to misalignment. Piping must also be secured and supported to prevent lateral movement which can create stress and damage the flange.



Proper Alignment **Improper Alignment**

2. With full face gasket in place, align the bolt holes of the mating flanges by rotating the ring into position with a Van Stone style flange or by rotating the pipe assembly with a one-piece flange installed.
3. Insert all bolts, washers (two standard flat washers per bolt), and nuts.
4. Make sure the faces of the mating surfaces are flush against gasket prior to bolting down the flanges.
5. Tighten all nuts by hand until they are snug. Establish uniform pressure over the flange face by tightening the bolts in 5 ft.-lb. increments according to the torque values shown in the following table using a 180° opposing sequence. Make sure that there are at least two (2) exposed threads beyond the nut upon final tightening.

NOTE: Thermoplastic materials relax over time. Re-torquing may be necessary 24 hours after initial tightening.

Flange Connections to other Equipment

For flange connections assembled to raised face flanges, appurtenances and/or equipment such as flow meters, expansion joints or wafer style butterfly valves where the flange face is not supported in direct contact with the mating flange, care must be taken to avoid "bending" the flange. Do not use bolts to bring together improperly mated flanges. Listed bolt torque may cause deformation or cracking from these types of connections since the flange is not fully supported by the mating flange. For flange connections to other equipment start with approximately two-thirds (2/3) of the listed maximum torque and increase as necessary to make the system leak-free after pressure testing.

Important: Be sure to contact the appurtenance or equipment manufacturer for recommendations regarding use of plastic flanges with their products.

Recommended Flange Bolt Torque for Plastic Flanges

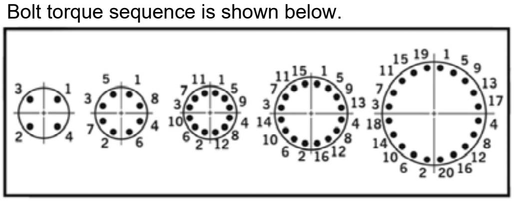
Flange Size (in.)	No. of Bolt Holes	Bolt Dia. (in.)[1]	Min. Bolt Length (in.)[1]	Torque ft.-lb.[2]
1/2	4	1/2	2	12
3/4	4	1/2	2	12
1	4	1/2	2-1/4	12
1-1/4	4	1/2	2-1/4	12
1-1/2	4	1/2	2-1/2	12
2	4	5/8	3	25
2-1/2	4	5/8	3-1/4	25
3	4	5/8	3-1/4	25
4	8	5/8	3-1/2	25
6	8	3/4	4	40
8	8	3/4	4-1/2	40
10	12	7/8	5	64
12	12	7/8	5	95
14	12	1	6	110
16	16	1	6-1/2	110
18	16	1-1/8	6-1/2	110
20[3]	20	1-1/8	5-1/2	110
24[3]	20	1-1/4	5-1/2	110

Note:

1. Minimum bolt length is based on connecting two (2) Spears® flanges, two (2) flat washers, gasket and nut. Adjustments will need to be made to accommodate valves and other equipment.
2. Bolt torque values based on lubricated bolts & nuts.
3. Bolt Length for Spears® Fabricated 20 inch & 24 inch Flanges with Plastic Rings.

Torque Sequence - Table 1

Bolt torque sequence is shown below.



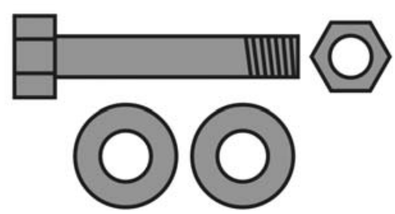
Temperature Pressure Rating

System Operating Temperature °F (°C)		100 (38)	110 (43)	120 (49)	130 (54)	140 (60)	150 (66)	160 (71)	170 (77)	180 (82)	190 (88)	200 (93)
Flange Pressure Rating psi (MPa)	PVC	150 (1.03)	135 (.93)	110 (.76)	75 (.52)	50 (.34)	-0- (-0-)	-0- (-0-)	-0- (-0-)	-0- (-0-)	-0- (-0-)	-0- (-0-)
	CPVC	150 (1.03)	140 (.93)	130 (.90)	120 (.83)	110 (.76)	100 (.70)	90 (.62)	80 (.55)	70 (.48)	60 (.41)	50 (.34)
	PP	150 (1.03)	105 (.72)	90 (.62)	80 (.55)	65 (.45)	50 (.34)	45 (.31)	30 (.21)	20 (.14)	-0- (-0-)	-0- (-0-)

Bolt Kit Selection Guide

Bolt Hardware Kits Available For Connection of 2-Spears® Flanges Includes Bolts, Nuts & Flat Washers for Specified Flange Size

- Order Gaskets & Bolt Kits Separately
- Pre-coated, Anti-seize Lubricated Bolts
- Available in Zinc Coated Steel, Type 316 Stainless Steel or Type 304 Stainless Steel



Flange Size	Bolts* Per Kit	Diameter (in.-TPI)	Length (in.)	Zinc	Kit Part Number 316 SSc	Kit Part Number 304 SSc
1/2	4	1/2 - 13	2	HK-005	HK1-005	HK2-005
3/4 & 1	4	1/2 - 13	2-1/4	HK-010	HK1-010	HK2-010
1-1/4 & 1-1/2	4	1/2 - 13	2-1/2	HK-015	HK1-015	HK2-015
2	4	5/8 - 11	3	HK-020	HK1-020	HK2-020
2-1/2	4	5/8 - 11	3-1/4	HK-025	HK1-025	HK2-025
3	4	5/8 - 11	3-1/2	HK-030	HK1-030	HK2-030
4	8	5/8 - 11	3-1/2	HK-040	HK1-040	HK2-040
6	8	3/4 - 10	4	HK-060	HK1-060	HK2-060
8	8	3/4 - 10	4-1/2	HK-080	HK1-080	HK2-080
10 & 12	12	7/8 - 9	6	HK-120	HK1-120	HK2-120

* Each Bolt Includes Nut & Two (2) Flat Washers

Configuration Terminology

Multi-Bolt Pattern Ring - Bolt hole drilling accepts ANSI and Metric Flanges
Socket - Slip socket connection for solvent cement welding
Spigot - Pipe O.D. connection for solvent welding
IPS - Iron Pipe Size

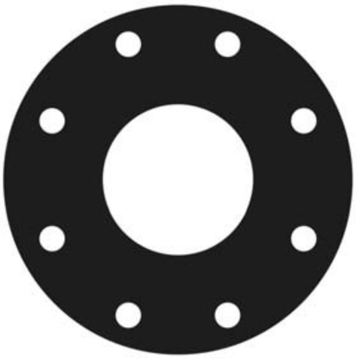
Gaskets

Full faced, 1/8" thick elastomer gaskets with a Shore "A" Durometer of approximately 70 is recommended.

Gasket Selection Guide

Following Gasket Numbers Available from Spears®

- Order Gaskets & Bolt Kits Separately
- 1/8" Full-Face design with ANSI Class 150 Bolt Patterns
- Pressure rated to 150 psi @ 73°F
- Available in Buna-N, EPDM, or FKM



Flange Size	Bolts Per Kit	Gasket Part Number		
		Buna-N	EPDM	FKM
1/2	4	GK1-005	GK2-005	GK3-005
3/4	4	GK1-007	GK2-007	GK3-007
1	4	GK1-010	GK2-010	GK3-010
1-1/4	4	GK1-012	GK2-012	GK3-012
1-1/2	4	GK1-015	GK2-015	GK3-015
2	4	GK1-020	GK2-020	GK3-020
2-1/2	4	GK1-025	GK2-025	GK3-025
3	4	GK1-030	GK2-030	GK3-030
4	8	GK1-040	GK2-040	GK3-040
6	8	GK1-060	GK2-060	GK3-060
8	8	GK1-080	GK2-080	GK3-080
10	12	GK1-100	GK2-100	GK3-100
12	12	GK1-120	GK2-120	GK3-120

CPVC Flange Sample Engineering Specification

All EverTUFF® CPVC flanges shall be produced by Spears® Manufacturing Company from CPVC materials, with a cell classification of 23447 in accordance with ASTM Standard D 1784. All injection-molded marine flanges shall be designed and manufactured to meet CL150 bolt pattern per ANSI Standard B16.5 and be certified by NSF International for use with potable water service. M851 and M854 series hubs shall accept ASTM IPS pipe. Rings shall be manufactured to accept bolt patterns in ANSI Class 150, B.S. 4504 PN10, B.S. 4504 PN16, B.S. 10 Table E for size 10". Metric Duetsche Industrie Norm (DIN) flange hubs shall accept 3" or 4" ASTM IPS Pipe. Rings shall be manufactured to DIN 8063-4 PN 10 and PN 16 bolt patterns. CPVC CL300 flanges shall accept ASTM IPS Pipe. CL300 rings shall be manufactured to the Class 300 bolt hole pattern from ANSI B16.5 and shall be pressure rated to 150psi @ 73° F.

Joining Method - Mechanical Grooved Couplings

In many installations where transition to metal pipe, or where disassembly is a prime factor, metallic grooved style couplings with gasket seal can be used to join CPVC pipe to alternate IPS size piping materials. In addition to the ease of disassembly, this type of connection also allows for a certain degree of angular adjustment and expansion/contraction. Special rolled-groove pipe can be joined, but easy to use molded Grooved Coupling Adapters then can be solvent cemented to plain end pipe are available for use with metallic grooved couplings.

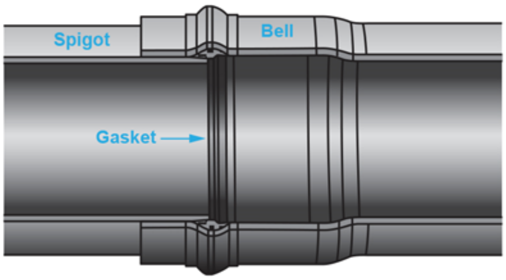
Only flexible style metallic grooved couplings are recommended for use with plastic pipe. Rigid style couplings should not be used as these can provide a compressive/shear load to plastic pipe resulting in failure. Always check the compatibility of the grooved coupling gasket material with the intended application fluids.

•NOTE A gasket/joint lubricant is recommended to prevent pinching the gasket and to assist the seating and alignment processes during assembly of grooved couplings. Certain lubricants may contain a petroleum base or other chemicals, which will cause damage to the plastic pipe, gasket and adapter. Always verify the suitability for use of the selected lubricant with the lubricant manufacturer.

Joining Method - Gasketed Pipe

Standards and Specifications

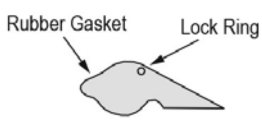
Since integral gasket bells are commercially available on a variety of pipe dimensions, applicable standards are dependent on the pipe dimension chosen. PVC pipe is manufactured from a Type I, Grade I PVC material per ASTM D 1784. Gasket pipe utilizes flexible elastomeric seals for pressure pipe. When properly assembled these joints meet the requirements of push-on joints per ASTM D 3139. The gaskets used should be manufactured in strict compliance with ASTM F477 requirements. SDR Series gasketed pipe is manufactured to ASTM D 2241. PVC Schedule 40, 80 and 120 gasketed pipe is manufactured to ASTM D 1785.



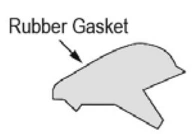
Typical Gasket Joint

Gasket Design

Gasketed pipe utilizes either gaskets that are locked in place at the factory as part of the manufacturing process, or insertable elastomer gaskets. Two styles of factory- installed gaskets are typically used. The Rieber style gasket with internally molded-in metal ring prevents fish mouthing or dislocation of the seal during assembly and the Ring style gasket. The standard gasket material typically used for both factory-installed gasket systems is Styrene Butadiene Rubber (SBR) which offers excellent physical properties and good chemical resistance. Other gasket materials are commercially available to meet demanding chemical resistance requirements.



Rieber Gasket - 2" - 24"
IPS & PIP Sizes
Laying Length - 19' 6" ± 2"



Ring Gasket - 18" - 27"
PIP Sizes Only
Laying Length - 19' ± 2"

Gasketed pipe offers low assembly force; flexibility to allow for variations in line pressure and changing working conditions; compensation for movement due to thermal expansion and contraction; a certain amount of allowable joint deflection; and positive, leak-free seals for both high- and low-pressure applications as well as vacuum service.

Field Assembly of Gasketed Joint

PVC pipes are assembled in the field by following a few simple steps, as outlined below:

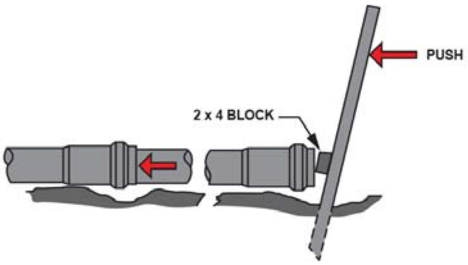
Step 1: Pipe end should have 15° bevel. Field cut pipe must be cut square and a 15° bevel applied to the cut spigot end. Factory pipe ends may have an insertion depth guide mark. If not, or if field cut, measure bell depth and mark on spigot end for insertion reference.

Step 2: Check that gasket is properly positioned in the bell groove. Clean the beveled end of the spigot and the gasket in the bell groove to be sure they are free of any particles or debris.

Step 3: Apply a lubricant recommended for PVC to the spigot end. Avoid applying lubricant directly on the gasket in the bell instead of the spigot.

WARNING: Some lubricants, oils or greases may not be compatible with PVC and can cause stress cracking. Verify PVC compatibility with the lubricant manufacturer.

Step 4: Insert and push lubricated spigot past the gasket into the bell until the guide mark on the spigot meets the end of the bell. It is usually possible to manually insert the spigot into the bell on pipe sizes 3" and smaller. Mechanical assistance may be needed for insertion on larger sizes. The "bar and block method" can be used, which allows the installer to feel the amount of force being used and whether the joint goes together smoothly.



"Bar and Block" Method for Mechanical Insertion of Spigot Into Bell

Deflection

Gasketed joints permit an angular deflection of 2° at the joint. Adequate deflection can usually be achieved for gentle curves by using the inherent flexibility of the pipe itself, without using joint deflection.

Thrust Blocking Gasket Joints

All gasket-joint piping requires adequate thrust restraints to prevent movement from forces generated by changes in direction, valve operation, dead ends, reduction in pipe size, and other areas where thrusts can be developed. The size and type of thrust restraint depends on the pipe size, type of fitting, soil properties, and water- hammer possibilities. Keeping flow velocities at or below 5 ft/sec will help minimize surge pressures. Fittings and valves used to make vertical changes in direction should be anchored to the thrust restraint to prevent outward and upward thrusts at the fitting junctures. In pressure lines, valves 3" in diameter and larger should be anchored to the thrust restraint to prevent movement when operated. Consideration should also be given for the proper support, anchoring, and thrust restraint for lines installed on slopes.

The size of thrust block required (in square feet) can be determined by dividing the total thrust developed (in psi) by the capacity of the soil (in pounds/square foot).

The most common method of thrust blocking involves the pouring of concrete (to the size of block required) between the pipe fitting and the bearing wall of the trench. Mechanical thrust restraint with PVC pipe.

Thrust in lb. @ 100 psi Operating Pressure

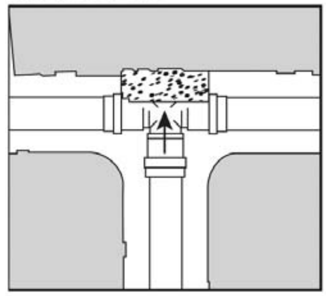
Pipe Size (in.)	90° Bend	45° Bend	22.5° Bend	Tee, Cap Plug, 60° Bend
2	645	350	180	455
2-1/2	935	510	260	660
3	1,395	755	385	985
4	2,295	1,245	635	1,620
6	4,950	2.680	1,370	3,500
8	8,375	4,540	2,320	5,930
10	13,040	7,060	3,600	9,230
12	18,340	10,000	5,240	13,000
14	21,780	11,770	6,010	15,400
16	28,440	15,370	7,850	20,110
18	35,990	19,450	9,930	25,450
20	44,430	24,010	12,260	31,420
24	63,970	34,570	17,650	45,240

Safe Bearing Capacity

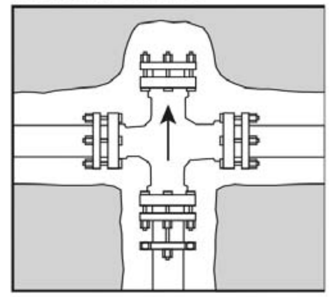
Soil	Capacity (lb./sq. ft.)
Muck, peat, etc.	0
Soft clay	1,000
Sand	2,000
Sand and gravel	3,000
Sand and gravel cemented with clay	4,000
Hard shale	10,000

Thrust Blocks

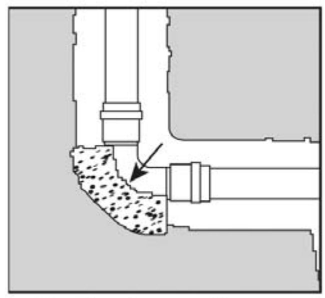
Thrust Retainers



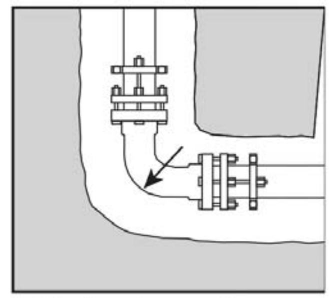
Thru Line connection, tee



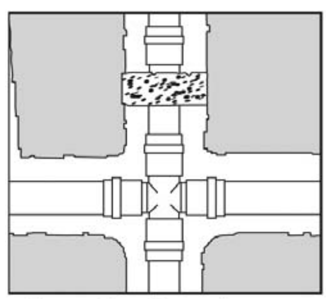
Thru Line connection, cross used as tee



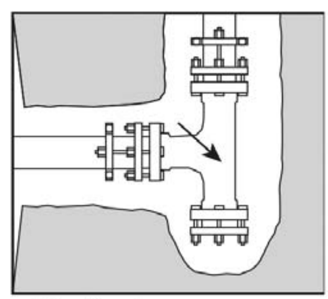
Direction change, elbow



Direction change, elbow



Change line size, reducer



Direction change, tee used as elbow

Assembly Instructions

Step One: Make certain pipe ends and gasket areas are free of dirt and debris. Support spigot end of pipe above ground to prevent dirt contamination when lubricant is applied.

Step Two: Apply a light coating of recommended lubricant to spigot end and sealing section of gasket.

WARNING - Use only lubricants specifically designated for use with PVC pipe. Certain greases, oils and vegetable oils will cause stress cracking in PVC materials.

Step Three: Align pipe ends. Push spigot end into gasket bell so that the reference mark is even with the entrance of the gasket bell.

Pounds of Force Required to Assemble Gasket Pipe

Rieber		Retaining Ring	
Pipe Size (in.)	ft.-lb.	Pipe Size (in.)	ft.-lb.
2	113	10	250
2-1/2	124	12	300
3	137	14	385
4	157	16	360
6	284	18	450
8	352	20	520
		24	600

Estimated Gasket Pipe Lubricant Use

Nominal Pipe Size (in.)	Avg. Number of Joints Per Pint (1 lb.) Container of lubricant
2	70
2-1/2	60
3	50
4	35
6	20
8	14
10	10
12	7
14	5
16	3
18	2
20	1.5
24	1

Trenching: Initial Backfill

Trench depth is determined by the intended service and local conditions. Gasket pipe should be buried a minimum of 12" below frost line in areas subject to freezing, or a minimum depth of 18" - 24" where there is no frost. Permanent lines subjected to heavy traffic should have a minimum cover of 24". In areas not subject to freezing, a minimum cover of 12" to 18" is usually sufficient for small-diameter piping subjected to light traffic. Bearing stresses must be calculated to determine the amount of cover required. Reference to applicable local, state, or national codes is also recommended.

The trench bottom should be continuous, relatively smooth, and free of rocks and debris. Adequate backfill should be in place immediately after installation, prior to filling or testing the line, to help distribute the effects of expansion/contraction evenly over each pipe length. The initial backfill material should consist of particles of 1/2" in size or less, and properly tamped. Generally a minimum of 6" - 12" of backfill is desirable for the initial phase. Where hardpan, ledge rock, or large boulders are encountered, the trench bottom should be padded with sand or compacted fine-grain soils to provide adequate protection. Joints should be left exposed for visual inspection during testing. Testing should be done before final backfill.

Testing

If separate tests are to be conducted for pressure and leakage, pressure testing should be conducted first.

WARNING: Air must be completely vented from the line prior to pressure testing; entrapped air can generate excessive surge pressures that are potentially damaging and can cause bodily injury or death. Air relief valves should be provided.

Section of pipe should be tested as it is installed to verify proper installation and joint assembly. Make certain the section of piping to be tested is backfilled sufficiently to prevent movement under test pressure. If concrete thrust blocks are utilized, allow sufficient time for concrete to set up prior to testing. Test ends must be capped and braced properly to withstand thrusts developed during testing. The following table provides the water volume requirements of various sizes of schedule and SDR series pipe.

Water Volume Gallons / 100'

Pipe Size (in.)	Sch. 40	Sch. 80	Sch. 120	SDR. 21	SDR. 26	SDR. 41
2	17	15	14	19	20	-
3	38	34	32	41	43	-
4	66	60	54	68	70	-
6	150	135	123	146	152	-
8	260	237	-	248	258	-
10	409	373	-	-	401	-
12	582	528	-	-	565	-
14	703	637	-	-	681	-
16	917	836	-	-	889	-
18	-	1060	-	-	1125	1195
20	-	-	-	-	1390	1475
24	-	-	-	-	2000	2125

Final Backfill

Backfilling should be conducted in layers; each layer must be compacted sufficiently so that lateral soil forces are developed uniformly. Under certain conditions it may be desirable to pressurize line during the backfill operation. Vibratory methods are recommended when compacting sand or gravel. Sand and gravel containing a significant proportion of fine-grained materials (silt, clay, etc.) should be compacted by mechanical tampers. When water flooding is used, sufficient cover must be provided by the initial backfill to ensure complete coverage of the pipe; precautions must be taken to prevent "floating" the pipe in the trench. Additional layers of backfill should not be applied until the water flooded backfill is firm enough to walk on. In all cases, the backfill should be placed and spread in uniform layers to eliminate voids. Large rocks, frozen dirt clods, and other debris larger than 3" should be removed to prevent damage to the pipe. Rolling equipment or heavy tampers should only be used to consolidate the final backfill. Additional information pertaining to underground installation is contained in ASTM D 2774 (Underground Installation of Thermoplastic Pressure Pipe), and ASTM D 2321 (Underground Installation of Flexible Thermoplastic Sewer Pipe).

Transition Joints & Specialty Fittings

PVC and CPVC pipe can be connected to steel, copper, brass, other metals and other plastic materials using a variety of transition fittings including unions, compression fittings, special reinforced adapters, flanged joints and grooved mechanical coupling joints.

Do not use regular PVC or CPVC female threaded fittings for connection to metal male threads. Spears® Special Reinforced (SR) female plastic threaded fittings are excellent for plastic to metal transitions. Unlike conventional plastic female adapters, these fittings incorporate the use of a stainless steel restraining collar located on the exterior of the FIPT threads of the adapter. This design allows direct connection to male metal threads without the need for pressure de-rating normally associated with conventional FIPT adapters, as the radial stress generated by thread engagement is contained. Other PVC/CPVC adapter fittings with brass or steel threads can also be used for transition to metal male threads.

If regular non-reinforced plastic threads must be transitioned to metal threads, the recommended joint is to use male plastic threads into female metal threads.

Underground Installation

Underground piping must be installed in accordance with any applicable codes. Attention should be given to local pipe laying techniques applicable to area subsoil. This may provide insights to particular pipe bedding issues. The following information is applicable to solvent cement joining of PVC and CPVC piping as a general guide. Refer to Gasketed Pipe section for additional information on installation of gasketed pipe.

Inspection: Before installation, PVC and CPVC piping products should be inspected for cuts, scratches, gouges or split ends. Damaged sections found must be cut-out and discarded.

Trenching: The trench should be as narrow as possible while providing adequate width to allow convenient installation. Minimum trench widths may be utilized by joining pipe outside of the trench and lowering it into the trench after adequate joint strength has been achieved. Refer to solvent cement instructions for recommended joint set and cure time.

Trench widths will have to be wider where pipe is joined in the trench or where thermal expansion and contraction is a factor.

Thermoplastic pipe should ALWAYS be installed below the frost level according to local conditions. Pipe for conveying liquids susceptible to freezing should be buried no less than 12" below the maximum frost level. Permanent lines subjected to heavy traffic should have a minimum cover of 24". For light traffic 12" to 18" is normally sufficient for small diameter pipe (typically < 3" diameter). With larger sizes, bearing stresses should be calculated to determine cover required.

When it is installed beneath surfaces that are subject to heavy weight or constant traffic such as roadways and railroad tracks, thermoplastic piping should be run within a metal or concrete casing. Refer to Critical Collapse Pressure Ratings for additional information.

The trench bottom should be continuous, relatively smooth and free of rocks. Where ledge rock, hardpan or boulders are encountered, it is necessary to pad the trench bottom using a minimum of four (4) inches of tamped earth or sand beneath the pipe as a cushion and for protection of the pipe from damage.

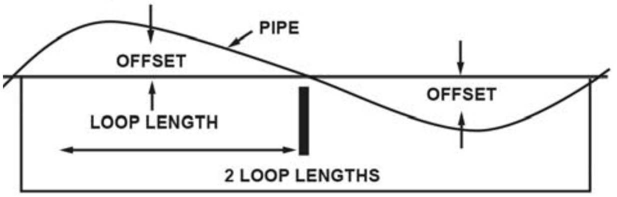
Sufficient cover must be maintained to keep external stress levels below acceptable design stress. Reliability and safety of service is of major importance in determining minimum cover. Local, state and national codes may also govern.

Snaking of Pipe may be used for small diameter piping systems (typically < 3"), but may also apply to larger diameter piping under specific applications and site conditions. Snaking of pipe is used to compensate for thermal expansion and contraction due to temperature changes. Snaking is particularly necessary on piping solvent welded during the late afternoon or a hot summer's day where drying time will extend through the cool of the night where thermal contraction could result in joint pull out. This snaking is also especially necessary with pipe that is laid in and is backfilled with cool earth before the joints are thoroughly dry.

After the pipe has been solvent welded and allowed to set properly, snake the pipe beside the trench during its required cure time. **BE ESPECIALLY CAREFUL NOT TO APPLY ANY STRESS THAT WILL DISTURB THE UNCURED JOINT.**

For Pipe Diameters < 3" diameter

Loop Offset in Inches for Contraction:



Maximum Temperature Variation, °F, Between Time of Solvent Welding and Final Use

Loop Length	10°	20°	30°	40°	50°	60°	70°	80°	90°	100°
LOOP OFFSET										
20	3"	4"	5"	5"	6"	6"	7"	7"	8"	8"
50	7"	9"	11"	13"	14"	16"	17"	18"	19"	20"
100	13"	18"	22"	26"	29"	32"	35"	37"	40"	42"

Expansion and contraction can be excessive in systems operating at near or at the maximum allowable temperature ranges with intermittent flow and buried lines. In these cases the lines should not be snaked. The use of properly installed expansion joints within a suitable concrete enclosure can be used. A section of larger diameter PVC pipe or other suitable sleeve should be used over the carrier pipe to pass through the wall of the concrete to minimize the potential for damage (scratching & scarring) as the result of movement caused by thermal expansion/contraction. Expansion joints should be suitably anchored independently of the carrier line. Axial guides should be used to direct movement into the expansion joint.

Backfilling: Underground pipe should be inspected and tested for leaks prior to backfilling. In hot weather, it is best to backfill early in the morning when the line is fully contracted and there is reduced chance of insufficiently dried joints being subject to contraction stresses.

The pipe should be uniformly and continuously supported over its entire length on firm, stable material. Blocking should not be used to change pipe grade or to intermittently support pipe across excavated sections.

Pipe is installed in a wide range of subsoils that must be stable and applied so as to physically shield the pipe from damage. Follow local pipe laying experience that may indicate particular pipe bedding problems.

Surround the pipe with 6" or 8" of backfill materials free of rocks and having a particle size of 1/2" or less. It should be placed in layers. Each soil layer should be sufficiently compacted to uniformly develop lateral passive soil forces during the backfill operation. It may be advisable to pressurize the pipe to 15 - 25 psi during the backfilling.

Vibratory methods are preferred when compacting sand or gravels. Best results are obtained when the soils are in a nearly saturated condition. Where water flooding is used, the initial backfill should be sufficient to ensure complete coverage of the pipe. Additional material should not be added until the water flooded backfill is firm enough to walk on. Care should be taken to avoid floating the pipe.

Sand and gravel containing a significant proportion of fine-grained material, such as silt and clay, should be compacted by hand or, preferably by mechanical tamper.

The remainder of the backfill should be placed and spread in uniform layers to fill the trench completely so that there will be no unfilled spaces under or around rocks or lumps of earth in the backfill. Remove large or sharp rocks, frozen clods and other debris greater than 3" in diameter. Rolling equipment or heavy tampers should only be used to consolidate the final backfill.

Avoid threaded connections in underground applications. Where transition to alternate materials is required the use of a flange component with suitable gasket is recommended. At vertical transitions from below ground systems to connections above ground, follow above ground installation procedures with regard to compensating for thermal expansion/contraction, weatherability, and proper support recommendations. Valves and other concentrated weight loads should be independently supported. Avoid excessive bending of pipe; excessive deflection of pipe and joints can reduce pressure bearing capability and cause failure.

Additional information on underground installations is contained in ASTM D 2774 "Underground Installation of Thermoplastic Pressure Piping", ASTM F 645, Standard Guide For "Selection Design and Installation of Thermoplastic Water Pressure Piping Systems", and ASTM D 2321 "Underground Installation of Thermoplastic Pipe for Sewers and Other Gravity-Flow Applications."

Above Ground Installation

Thermal Expansion & Contraction

Attention must be given to above ground installations where ambient temperature swings can cause thermoplastic systems to expand and contract both indoors and out. For example, a system installed in an unheated building during the winter months will expand considerably when temperatures rise. Conversely, systems installed at higher ambient temperatures will contract as temperatures fall. Refer to Thermal Expansion & Contraction section for additional information.

Outdoor Applications & Protection

PVC and CPVC piping systems must be protected from freezing. Many standard cold weather piping design and installation practices can be used to protect the system from freezing, such as use of pipe insulation, anti-freeze solutions, and heat trace tapes. The suitability and compatibility of these products for use with PVC and CPVC should be verified with product manufacturers prior to use.

Caution should be exercised in installing PVC and CPVC piping products in metal boxes or enclosures exposed to direct sunlight. Such enclosures can act as "ovens" that significantly increase the environmental temperature over ambient air conditions, resulting in product damage and failure.

PVC and CPVC piping exposed to the direct sunlight (UV radiation) should be painted with a reflective, light colored acrylic or latex paint. Avoid dark colors, especially black. Heat absorption can exceed the heat handling capacity of the pipe and fitting material. Compatibility information regarding use with PVC/CPVC products should be confirmed with the paint manufacturer. Oil-based paints should not be used.

Hangers and Supports

Hanger Support Spacing

Support location and spacing depends on the pipe diameter, system operating temperature, and the location of any concentrated stress loads (i.e., valves, flanges, test equipment and any other heavy system components). Hangers must have an adequate load-bearing surface free of any rough or sharp edges that could damage the pipe during use. Hangers also must not restrict linear movement of the system in order to allow thermal expansion and contraction from temperature changes.

Proper support spacing can be calculated similarly to that of metal systems by using simple and continuous beam calculations. This can be achieved using the maximum fiber stress of the material, or deflection based on the long term modulus of the material at the temperature selected as the limiting factors.

Hanger Selection

Hangers designed for metallic pipe can be used if they provide an adequate load-bearing surface, which is smooth and free of rough or sharp edges that could damage the pipe. Improper supports can generate excessive sag resulting in failure. Movement caused by thermal expansion/contraction and pressure fluctuations must be considered. Hangers and supports used must permit axial movement of the system, but not compress the pipe. Supplemental guides may be required in addition to hangers in order to maintain alignment and direct movement into in-line Expansion Joints.

Placement

Hangers should be installed within two feet of each side of a pipe joint; while changes in direction should be supported as close as possible to the fitting to reduce stress. Heavy system components such as valves, flanged assemblies, tees and other concentrated stress loads must be independently supported. Valves should additionally be adequately braced to prevent movement/stress loads from operational torque. Support of potential solids accumulation loads within the line should also be considered.

Precautions

Protective sleeves or pads used between the pipe and the hanger will distribute stress loads over a greater surface area, especially when using U-bolt or roller type hangers. Protective sleeves or pads should also be used when horizontal pipe is resting on concrete or other abrasive support structures. Do not allow piping to contact abrasive surfaces that could cause damage during axial movement. Avoid contact with heat producing sources. Do not install plastic piping in close proximity to steam lines or other high temperature equipment without providing appropriate protection to prevent damage from distortion or expansion/contraction. Care should be taken to avoid over tightening of anchors, clamps or other support devices. This can distort or even fracture the piping. Vertical lines must be supported properly at intervals that will prevent excessive loading on the fitting at the lower end. Hangers and clamps suitable for this purpose include riser clamps or double bolt type clamps installed in such a manner that will allow for movement of the pipe due to thermal expansion and contraction.

Clamps and hangers used must not compress, distort, cut or abrade the piping. Common practice is to install clamps just below a coupling so that the shoulder of the coupling rests on the clamp. Fittings can be modified in the field to achieve this by cutting a coupling in two, just above the stop at the socket bottom, and then cutting this piece in half lengthwise to provide two halves which do not contain the stop. The two halves are then solvent cemented to the pipe at the proper location so that the shoulder of the modified coupling rests on the clamp. Riser clamps that utilize compression to support the pipe weight should not be used.

Anchor Guides

Anchors direct movement of the piping by providing restraint at key points such as long straight runs, at changes in direction of the system, and where expansion joints and other methods of thermal compensation are utilized. They may be used to control forces caused by expansion and contraction, generated by pressure surges, vibration, and other transient conditions. Guides are necessary to help direct this movement between anchors by allowing longitudinal movement while restricting lateral movement. Depending on the application and type, guides may or may not act as supports. Support guides should have the same load bearing surface and other requirements of hangers designed for the system. Guides must be rigidly attached to the structure to prevent lateral movement, but should not restrict longitudinal movement of the pipe through the guide. Anchors and guides must be engineered and installed without point loading the system.

Recommended Pipe Support Spacing

The following hanger support spacing recommendations are according to size, schedule, and operating temperatures. Do not clamp supports tightly - this restricts axial movement of the pipe. If short spacing is necessary, continuous supports may be more economical. These are considered conservative in nature and are based on straight runs of un-insulated lines with fluids with a specific gravity of 1.00 or less. These values do not consider concentrated weight loads or aggressive reagents.

Support Spacing Adjustment for Fluids of Different Specific Gravity

Recommendations for PVC and CPVC piping support spacing in the following table are based on straight runs of uninsulated lines conveying fluids with specific gravities up to 1.0 (water). For specific gravities greater than 1.0, the spacing from the Support spacing tables should be multiplied by the following correction factors. System components such as valves, flanged assemblies, tees and other forms of concentratred loads must be independently supported.

Specific Gravity	1.0	1.1	1.2	1.4	1.6	2.0	2.5
Correction Factor	1.00	.98	.96	.93	.90	.85	.80

Temp (F)	1/2"	3/4"	1"	1-1/4"	1-1/2"	2"	2-1/2"	3"	4"	6"	8"	10"	12"	14"	16"
73	5-1/2	5-1/2	6	6-1/2	7	7	8	8	9	10	11	11-1/2	12-1/2	15	16
100	5	5-1/2	6	6	6-1/2	7	7-1/2	8	9	9-1/2	10-1/2	11	12-1/2	13-1/2	15
120	4-1/2	5	5-1/2	6	6	6-1/2	7-1/2	7-1/2	8-1/2	9	10	10-1/2	11	12-1/2	13-1/2
140	4-1/2	4-1/2	5	5-1/2	5-1/2	6	6-1/2	7	7-1/2		9	9-1/2	10-1/2	11	12
160	3	3	3-1/2	3-1/2	3-1/2	4	4-1/2	4-1/2	5	5-1/2	6	6-1/2	7-1/2	9-1/2	10
180	2-1/2	2-1/2	3	3	3-1/2	3-1/2	4	4	4-1/2	5	5-1/2	6	6-1/2	8	8-1/2

Suitable for Oil-Free air handling to 25 psi, not for distribution of compressed air or gas

See Spears® Product Sourcebook for product offerings

PVC PIPE SUPPORT SPACING (ft.)

PIPE SIZE (in.)	SCHEDULE 40					SCHEDULE 80					SCHEDULE 120				
	60°F	80°F	100°F	120°F	140°F	60°F	80°F	100°F	120°F	140°F	60°F	80°F	100°F	120°F	140°F
1/4	4	3-1/2	3-1/2	2	2	4	4	3-1/2	2-1/2	2	---	---	---	---	---
3/8	4	4	3-1/2	2-1/2	2	4-1/2	4-1/2	4	2-1/2	2-1/2	---	---	---	---	---
1/2	4-1/2	4-1/2	4	2-1/2	2-1/2	5	4-1/2	4-1/2	3	2-1/2	5	5	4-1/2	3	2-1/2
3/4	5	4-1/2	4	2-1/2	2-1/2	5-1/2	5	4-1/2	3	2-1/2	5-1/2	5	4-1/2	3	3
1	5-1/2	5	4-1/2	3	2-1/2	6	5-1/2	5	3-1/2	3	6	5-1/2	5	3-1/2	3
1-1/4	5-1/2	5-1/2	5	3	3	6	6	5-1/2	3-1/2	3	6-1/2	6	5-1/2	3-1/2	3-1/2
1-1/2	6	5-1/2	5	3-1/2	3	6-1/2	6	5-1/2	3-1/2	3-1/2	6-1/2	6-1/2	6	4	3-1/2
2	6	5-1/2	5	3-1/2	3	7	6-1/2	6	4	3-1/2	7-1/2	7	6-1/2	4	3-1/2
2-1/2	7	6-1/2	6	4	3-1/2	7-1/2	7-1/2	6-1/2	4-1/2	4	8	7-1/2	7	7-1/2	4
3	7	7	6	4	3-1/2	8	7-1/2	7	4-1/2	4	8-1/2	8	7-1/2	5	4-1/2
3-1/2	7-1/2	7	6-1/2	4	4	8-1/2	8	7-1/2	5	4-1/2	9	8-1/2	7-1/2	5	4-1/2
4	7-1/2	7	6-1/2	4-1/2	4	9	8-1/2	7-1/2	5	4-1/2	9-1/2	9	8-1/2	5-1/2	5
5	8	7-1/2	7	4-1/2	4	9-1/2	9	8	5-1/2	5	10-1/2	10	9	6	5-1/2
6	8-1/2	8	7-1/2	5	4-1/2	10	9-1/2	9	6	5	11-1/2	10-1/2	9-1/2	6-1/2	6
8	9	8-1/2	8	5	4-1/2	11	10-1/2	9-1/2	6-1/2	5-1/2	---	---	---	---	---
10	10	9	8-1/2	5-1/2	5	12	11	10	7	6	---	---	---	---	---
12	11-1/2	10-1/2	9-1/2	6-1/2	5-1/2	13	12	10-1/2	7-1/2	6-1/2	---	---	---	---	---
14	12	11	10	7	6	13-1/2	13	11	8	7	---	---	---	---	---
16	12-1/2	11-1/2	10-1/2	7-1/2	6-1/2	14	13-1/2	11-1/2	8-1/2	7-1/2	---	---	---	---	---
18	13	12	11	8	7	14-1/2	14	12	11	9	---	---	---	---	---
20	14	12-1/2	11-1/2	10	8-1/2	15-1/2	14-1/2	12-1/2	9-1/2		---	---	---	---	---
24	15	13	12-1/2	11	9-1/2	17	15	14	12-1/2	10-1/2	---	---	---	---	---
	SDR 41					SDR 26									
18	13	12	11	8	7	14-1/2	14	12	9	8	---	---	---	---	---
20	13-1/2	12-1/2	11-1/2	8-1/2	7-1/2	15	14-1/2	12-1/2	9-1/2	8-1/2	---	---	---	---	---
24	14	13	12	9	8	15-1/2	15	13	10	9	---	---	---	---	---

•**NOTE** Although support spacing is shown at 140°F for PVC, consideration should be given to the use of CPVC or continuous support above 120°F. The possibility of temperature overrides beyond regular working temperatures and cost may make either of the alternatives more desirable. This chart based on continuous spans and for un-insulated line carrying fluids of specific gravity up to 1.00.

CPVC PIPE SUPPORT SPACING (ft.)

PIPE SIZE (in.)	SCHEDULE 40						SCHEDULE 80					
	73°F	100°F	120°F	140°F	160°F	180°F	73°F	100°F	120°F	140°F	160°F	180°F
1/2	5	4-1/2	4-1/2	4	2-1/2	2-1/2	5-1/2	5	4-1/2	4-1/2	3	2-1/2
3/4	5	5	4-1/2	4	2-1/2	2-1/2	5-1/2	5-1/2	5	4-1/2	3	2-1/2
1	5-1/2	5-1/2	5	4-1/2	3	2-1/2	6	6	5-1/2	5	31/2	3
1-1/4	5-1/2	5-1/2	5-1/2	5	3	3	6-1/2	6	6	5-1/2	31/2	3
1-1/2	6	6	5-1/2	5	3-1/2	3	7	6-1/2	6	5-1/2	31/2	3-1/2
2	6	6	5-1/2	5	3-1/2	3	7	7	6-1/2	6	4	3-1/2
2-1/2	7	7	6-1/2	6	4	3-1/2	8	7-1/2	7-1/2	6-1/2	41/2	4
3	7	7	7	6	4	3-1/2	8	8	7-1/2	7	41/2	4
3-1/2	7-1/2	7-1/2	7	6-1/2	4	4	8-1/2	8-1/2	8	7-1/2	5	4-1/2
4	7-1/2	7-1/2	7	6-1/2	4-1/2	4	8-1/2	9	8-1/2	7-1/2	5	4-1/2
6	8-1/2	8	7-1/2	7	5	4-1/2	10	9-1/2	9	8	51/2	5
8	9-1/2	9	8-1/2	7-1/2	5-1/2	5	11	10-1/2	10	9	6	5-1/2
10	10-1/2	10	9-1/2	8	6	5-1/2	11-1/2	11	10-1/2	9-1/2	61/2	6
12	11-1/2	10-1/2	10	8-1/2	6-1/2	6	12-1/2	12	11-1/2	10-1/2	71/2	6-1/2
14	12	11	10	9	8	6	15	13-1/2	12-1/2	11	91/2	8
16	13	12	11	9-1/2	8-1/2	7	16	15	13-1/2	12	10	8-1/2

•**NOTE** This chart based on continuous spans and for un-insulated line carrying fluids of specific gravity up to 1.00.

PVC PIPE SUPPORT SPACING (FT) AT °F (°C)

Nominal pipe size (in.)	SCHEDULE 40					
	20°F (-7°C)	60°F (16°C)	80°F (27°C)	100°F (38°C)	120°F (49°C)	140°F (60°C)
1/4	4.5	4.0	3.5	3.5	2.0	2.0
3/8	4.5	4.0	4.0	3.5	2.5	2.0
1/2	5.0	4.5	4.5	4.0	2.5	2.5
3/4	5.5	5.0	4.5	4.0	2.5	2.5
1	6.0	5.5	5.0	4.5	3.0	2.5
1 1/4	6.0	5.5	5.5	5.0	3.0	3.0
1 1/2	6.5	6.0	5.5	5.0	3.5	3.0
2	6.5	6.0	5.5	5.0	3.5	3.0
2 1/2	7.5	7.0	6.5	6.0	4.0	3.5
3	8.0	7.0	7.0	6.0	4.0	3.5
4	8.5	7.5	7.0	6.5	4.5	4.0
6	9.5	8.5	8.0	7.5	5.0	4.5
8	10.0	9.0	8.5	8.0	5.0	4.5
10	11.0	10.0	9.0	8.5	5.5	5.0
12	12.0	11.5	10.5	9.5	6.5	5.5
14	13.0	12.0	11.0	10.0	7.0	6.0
16	14.0	12.5	11.5	10.5	7.5	6.5

PVC PIPE SUPPORT SPACING (FT) AT °F (°C)

NOMINAL PIPE SIZE (IN.)	SCHEDULE 80					
	20°F (-7°C)	60°F (16°C)	80°F (27°C)	100°F (38°C)	120°F (49°C)	140°F (60°C)
1/4	4.5	4.0	4.0	3.5	2.5	2.0
3/8	5.0	4.5	4.5	4.0	2.5	2.5
1/2	5.5	5.0	4.5	4.5	3.0	2.5
3/4	6.0	5.5	5.0	4.5	3.0	2.5
1	6.5	6.0	5.5	5.0	3.5	3.0
1 1/4	7.0	6.0	6.0	5.5	3.5	3.0
1 1/2	7.0	6.5	6.0	5.5	3.5	3.5
2	7.5	7.0	6.5	6.0	4.0	3.5
2 1/2	8.5	7.5	7.5	6.5	4.5	4.0
3	9.0	8.0	7.5	7.0	4.5	4.0
4	10.0	9.0	8.5	7.5	5.0	4.5
6	11.0	10.0	9.5	9.0	6.0	5.0
8	12.5	11.0	10.5	9.5	6.5	5.5
10	13.5	12.0	11.0	10.0	7.0	6.0
12	14.5	13.0	12.0	10.5	7.5	6.5

PVC PIPE SUPPORT SPACING (FT) AT °F (°C)

NOMINAL PIPE SIZE (IN.)	SDR 13.5					
	20°F (-7°C)	60°F (16°C)	80°F (27°C)	100°F (38°C)	120°F (49°C)	140°F (60°C)
1/2	4.5	4.0	4.0	3.5	C	C
3/4	5.0	4.5	4.0	3.5	C	C
1	5.5	5.0	4.5	4.0	2.5	C
1 1/4	5.5	5.0	5.0	4.5	2.5	2.5
1 1/2	6.5	6.0	5.5	5.0	3.5	3.0
2	7.0	6.5	6.0	5.5	3.5	3.0
2 1/2	8.0	7.0	7.0	6.0	4.0	3.5
3	8.5	7.5	7.0	6.5	4.0	3.5
4	9.5	8.5	8.0	7.0	5.0	4.5

PVC PIPE SUPPORT SPACING (FT) AT °F (°C)

NOMINAL PIPE SIZE (IN.)	SDR-26					
	20°F (-7°C)	60°F (16°C)	80°F (27°C)	100°F (38°C)	120°F (49°C)	140°F (60°C)
1	3.0	2.5	2.5	2.5	C	C
1 1/4	3.0	2.5	2.5	2.5	C	C
1 1/2	3.0	3.0	2.5	2.5	C	C
2	3.0	3.0	2.5	2.5	C	C
2 1/2	3.5	3.5	3.0	2.5	2.0	C
3	5.0	4.0	4.0	3.5	2.5	2.0
4	6.0	5.5	4.5	4.0	3.0	2.0
5	6.0	5.5	5.0	4.5	3.0	2.5
6	6.5	6.0	5.5	5.0	3.5	3.0
8	7.0	6.0	6.0	5.5	3.5	3.0
10	7.5	7.0	6.0	6.0	4.0	3.5
12	8.5	8.0	7.5	6.5	4.5	4.0
14	9.0	8.5	7.5	7.0	5.0	2.5
16	10.0	9.0	8.5	7.5	5.5	4.0
18	11.0	9.5	9.0	8.0	6.0	4.5
20	12.0	10.0	9.5	8.5	6.5	5.0
24	13.0	11.0	10.0	9.0	7.0	5.5

Suitable for Oil-Free air handling to 25 psi, not for distribution of compressed air or gas
See Spears® Product Sourcebook for product offerings

PVC PIPE SUPPORT SPACING (FT) AT °F (°C)

NOMINAL PIPE SIZE (IN.)	SDR-21					
	20°F (-7°C)	60°F (16°C)	80°F (27°C)	100°F (38°C)	120°F (49°C)	140°F (60°C)
1/4	2.5	2.0	2.0	C	C	C
1	3.0	2.5	2.5	2.0	C	C
1 1/4	3.5	3.0	3.0	2.5	C	C
1 1/2	4.0	3.5	3.0	2.5	2.0	C
2	4.0	3.5	3.0	3.0	2.0	2.0
2 1/2	4.5	4.0	4.0	3.5	2.0	2.0
3	5.0	4.0	3.5	3.5	2.5	2.0
4	8.0	7.0	6.5	6.0	4.0	3.5
5	7.0	6.5	4.0	4.0	2.0	1.5
6	6.0	5.5	5.0	5.0	3.0	2.5
8	6.5	6.0	5.5	5.0	3.5	3.0
10	7.0	6.5	6.0	5.5	3.5	3.0
12	7.5	7.0	6.5	5.5	4.0	3.5

PVC PIPE SUPPORT SPACING (FT) AT °F (°C)

NOMINAL PIPE SIZE (IN.)	SDR-41					
	20°F (-7°C)	60°F (16°C)	80°F (27°C)	100°F (38°C)	120°F (49°C)	140°F (60°C)
4	3.5	3.0	2.5	2.5	C	C
6	4.0	3.5	3.0	3.0	2.0	C
8	4.0	3.5	3.5	3.0	2.0	C
10	4.5	4.0	3.5	3.5	2.5	2.0
12	5.0	5.0	4.5	4.0	2.5	2.5
14	5.5	5.0	4.5	4.0	3.0	2.0
16	6.0	5.0	5.0	4.5	3.0	2.5
18	6.5	6.0	5.5	5.0	3.5	2.5
20	7.5	6.0	6.0	5.0	4.0	3.0
24	8.0	6.5	6.5	5.5	4.0	3.0

CPVC PIPE SUPPORT SPACING (FT) AT °F (°C)

NOMINAL PIPE SIZE (IN.)	SCHEDULE 40						
	60°F (16°C)	100°F (38°C)	120°F (49°C)	140°F (60°C)	160°F (71°C)	180°F (82°C)	200°F (93°C)
1/4	4.5	4.0	3.5	3.5	2.0	2.0	C
3/8	4.5	4.0	4.0	3.5	2.5	2.0	C
1/2	5.0	4.5	4.5	4.0	2.5	2.5	C
3/4	5.5	5.0	4.5	4.0	2.5	2.5	C
1	6.0	5.5	5.0	4.5	3.0	2.5	C
1 1/4	6.0	5.5	5.5	5.0	3.0	3.0	1.5
1 1/2	6.5	6.0	5.5	5.0	3.5	3.0	1.5
2	6.5	6.0	5.5	5.0	3.5	3.0	1.5
2 1/2	7.5	7.0	6.5	6.0	4.0	3.5	1.5
3	8.0	7.0	7.0	6.0	4.0	3.5	2.0
4	8.5	7.5	7.0	6.5	4.5	4.0	2.0
5	9.5	8.5	8.0	7.5	5.0	4.5	2.0
6	9.5	8.0	7.5	7.0	5.0	4.5	2.0
8	10.0	9.0	8.5	7.5	5.5	5.0	2.0
10	10.5	10.0	9.5	8.0	6.0	5.5	2.0
12	11.0	10.5	10.0	8.5	6.5	6.0	2.5

CPVC PIPE SUPPORT SPACING (FT) AT °F (°C)

NOMINAL PIPE SIZE (IN.)	SCHEDULE 80						
	60°F(16°C)	100°F (38°C)	120°F (49°C)	140°F (60°C)	160°F (71°C)	180°F (82°C)	200°F (93°C)
1/4	4.5	4.0	4.0	3.5	2.5	2.0	C
3/8	5.0	4.5	4.5	4.0	2.5	2.5	C
1/2	5.5	5.0	4.5	4.5	3.0	2.5	C
3/4	5.5	5.5	5.0	4.5	3.0	2.5	C
1	6.0	6.0	5.5	5.0	3.5	3.0	1.5
1 1/4	6.5	6.0	6.0	5.5	3.5	3.0	1.5
1 1/2	7.0	6.5	6.0	5.5	3.5	3.5	2.0
2	7.0	7.0	6.5	6.0	4.0	3.5	2.0
2 1/2	8.0	7.5	7.5	6.5	4.5	4.0	2.5
3	8.0	8.0	7.5	7.0	4.5	4.0	2.5
4	8.5	9.0	8.5	7.5	5.0	4.5	2.5
5	10.0	9.5	9.0	8.0	5.5	5.0	3.0
6	10.0	9.5	9.0	8.0	5.5	5.0	3.0
8	11.0	10.5	10.0	9.0	6.0	5.5	3.5
10	11.5	11.0	10.5	9.5	6.5	6.0	4.0
12	12.5	12.0	11.5	10.5	7.5	6.5	4.5